

Sukhoi Su-34

The **Sukhoi Su-34** (Russian: Сухой Су-34; NATO reporting name: **Fullback**)^[7] is a Soviet-origin Russian twin-engine, twin-seat, all-weather supersonic medium-range fighter-bomber/strike aircraft.^{[8][9][10][11]} It first flew in 1990, intended for the Soviet Air Forces, and it entered service in 2014 with the Russian Air Force.^[12]

Based on the Sukhoi Su-27 Flanker air superiority fighter, the Su-34 has a wider, armoured cockpit with side-by-side seating for its two pilots.^[13] The Su-34 was designed primarily for tactical deployment against ground and naval targets (tactical bombing/attack/interdiction roles, including against small and mobile targets) on solo and group missions in daytime and at night, under favourable and adverse weather conditions and in a hostile environment with counter-fire and electronic warfare (EW) counter-measures deployed, as well as for aerial reconnaissance.^{[8][14]} The Su-34 is planned to eventually replace the Su-24 tactical bomber and the Tu-22M long-range bomber.^[13]

Development

Beginnings (1980s)

The Su-34 had a murky and protracted beginning.^[15] In the mid-1980s, Sukhoi began developing a new tactical multirole combat aircraft to replace the swing-wing Su-24, which would incorporate a host of conflicting requirements. The bureau thus selected the Su-27, which excelled in maneuverability and range, and could carry a large payload, as the basis for the new fighter-bomber.^[16] More specifically, the aircraft was developed from T10KM-2, the naval trainer derivative of the Sukhoi Su-27K. The development, known internally as T-10V, was shelved at the end of the 1980s sharing the fate of the aircraft carrier Ulyanovsk; this was the result of the political upheaval in the Soviet Union and its subsequent disintegration.^{[15][16]}

Su-34 / Su-32



A Russian Aerospace Forces Su-34 performing at the 2015 MAKS air show

General information

Type	<u>Fighter-bomber/strike fighter</u>
National origin	<u>Soviet Union/Russia</u>
Manufacturer	<u>Sukhoi, Novosibirsk Aircraft Production Association (NAPO)</u>
Status	<u>In service</u>
Primary users	<u>Russian Aerospace Forces</u> <u>Algerian Air Force</u>
Number built	>163 (7 test and 156 serial aircraft in early 2024) ^{[1][2][3]}

History

Manufactured	2006–present ^{[4][5]}
Introduction date	20 March 2014 ^[6]
First flight	13 April 1990
Developed from	<u>Sukhoi Su-27</u>

Prototypes (1990–1993)

In August 1990, a photograph taken by a TASS officer showed an aircraft making a dummy approach towards the aircraft carrier Tbilisi.^{[15][16]} The aircraft, subsequently and erroneously labelled Su-27KU by Western intelligence, made its maiden flight on 13 April 1990 with Anatoliy Ivanov at the controls.^[17] Converted from an Su-27UB with the new distinctive nose, while retaining the main undercarriage of previous Su-27s, it was a prototype for the Su-27IB (IB stands for *istrebitel-bombardirovshchik*, or "fighter bomber").^[18] It was developed in parallel with the two-seat naval trainer, the Su-27KUB. However, contrary to earlier reports, the two aircraft are not directly related.^[19] Flight tests continued throughout 1990 and into 1991.^[16]

In 1992, the Su-27IB was displayed to the public at the MosAeroshow (later renamed "MAKS Airshow"), where it demonstrated aerial refuelling with an Il-78, and performed an aerobatic display. The aircraft was officially unveiled on 13 February 1992 at Machulishchy (air base), where Russian president Boris Yeltsin and the Commonwealth of Independent States (CIS) leaders were holding a summit. The following year, the Su-27IB was again displayed at the MAKS Airshow.^[17]

The next prototype, and first pre-production aircraft, T10V-2, first flew on 18 December 1993, with Igor Votintsev and Yevgeniy Revoonov at the controls.^[17] Built at Novosibirsk, where Su-24s were constructed, this aircraft was visibly different from the original prototype; it had modified vertical stabilizers, twin tandem main undercarriage and a longer "stinger", which houses an APU in order to balance out the weight from the cockpit.^[15]

Standard Su-34 (1994)

The first aircraft built to production standard made its maiden flight on 28 December 1994.^[15] It was fitted with a fire-control system, at the heart of which was the Leninets OKB-designed V004 passive electronically scanned array radar.^[15] It was different enough from the earlier versions that it was re-designated the "Su-34".^[17] However, at the 1995 Paris Air Show, the aircraft was allocated the "Su-32FN" designation, signalling the aircraft's potential role as a shore-based naval aircraft for the Russian Naval Aviation. Sukhoi also promoted the Su-34 as the "Su-32MF" (*MnogoFunksionalniy*, "multi-function").^[15]



Su-32FN in Paris Airshow 1995

Budget restrictions caused the programme to stall repeatedly. Nevertheless, flight testing continued, albeit at a slow pace. The third pre-production aircraft first flew in late 1996.^[20]

Further development (2016–present)

Russia's Ministry of Defence planned to modernize the Su-34; according to the deputy head of the military department, Yuriy Borisov, "We are planning to modernize the aircraft: prolong its service life, increase the number of airborne weapons. Plane is in great demand in our armed forces, and it has a great future."^[21]

Russia developed two new versions of the aircraft: one for electronic warfare (L700 Tarantul ECM pod can provide electronic cover for a group of aircraft)^{[22][23]} and one for intelligence, surveillance, and reconnaissance.^{[24][25]}

Su-34M modernised version features a new electro-optical infrared targeting pod, a Kopyo-DL rearward facing radar that can warn the pilots if missiles are approaching, combined with automatic deployment of countermeasures and jamming.^[25]



An Su-34 over the Baltic in 2015, photographed while being intercepted by the Royal Air Force

Orders and deliveries

Su-34 (2004–2021)

An initial batch of eight aircraft was completed by the Novosibirsk factory in 2004.^[26] In March 2006, Russian Minister of Defence Sergei Ivanov announced the purchase of the first five pre-production Su-34s for the Russian Air Force. In late 2008, a second contract was signed for delivery of 32 aircraft by 2015.^[27] A total of 70 aircraft were to be purchased by 2015 to replace some 300 Su-24s in service at the time, which were then undergoing a modernization program.^[28] Ivanov claimed that as it is "many times more effective on all critical parameters", fewer of these newer bombers are required than the old Su-24 it replaces. In December 2006, Ivanov stated that approximately 200 Su-34s were expected to be in service by 2020. This was confirmed by Russian Air Force chief Vladimir Mikhaylov on 6 March 2007. Two Su-34s were delivered in 2006–2007, and three more were delivered by the end of 2009.^[27]

On 9 January 2008, Sukhoi reported that the Su-34 had begun full-rate production. The final stage of the state tests were completed on 19 September 2011.

The Russian Air Force received another four Su-34s on 28 December 2010, as combat units in airbases first received six Su-34s in 2011. Delivery came in the form of two contracts, the first in 2008 for 32 aircraft and the second in 2012 for a further 92 aircraft, totaling 124 to be delivered by 2020.^[29]

In December 2012, Sukhoi reportedly delivered five aircraft under the 2012 State Defence Order.^[30]



A Sukhoi Su-34 at MAKS-2013

On 6 May 2013, the first Su-34s under the 2013 defence procurement plan were delivered.^[31] On 9 July 2013, three more Su-34s were delivered in an official acceptance ceremony held at the Novosibirsk Aircraft Plant.^[32] By the end of 2013, Sukhoi completed the 2008 contract and started deliveries of aircraft under the 2012 contract.^[33]

In August 2013, Sukhoi ordered 184 identification friend or foe transponders for the Russian Su-34s from the Kazan-based Radiopribor holding company, to be delivered by 2020.

On 10 June 2014, a further delivery of Su-34s was made to the 559th Bomber Aviation Regiment at Morozovsk Air Base.^[34] Another three aircraft were delivered on 18 July 2014.^[35] In total, 18 aircraft were delivered in 2014,^{[36][37]} with 20 planned to be delivered in 2015.^[38]

Sukhoi delivered the first batch of Su-34s under the 2015 defence procurement plan on 21 May 2015.^[39] On 16 July 2015, the Sukhoi Company handed over another batch of Su-34 frontline bombers to the Russian Defence Ministry. The transfer took place in the framework of the Unified Military Hardware Acceptance Day in the Novosibirsk Aircraft Production Association Plant.^[40]

Algerian deal: after eight years of negotiations, Rosoboronexport received official request to supply the Su-34 to the Algerian Air Force.^[41]

The last two Su-34s of the 2012 contract were delivered to the 968th Fighter Aviation Regiment at the Lipetsk Air Base in December 2020. This brought the total number of operational aircraft delivered to the Russian Aerospace Forces to 131 (not counting crashes and aircraft written off). Apart from this, also seven pre-production units were built under previous contracts.^[42]

On 25 August 2020, the Russian Defence Ministry signed a third contract for a further 24 Su-34 aircraft for the Russian Aerospace Forces. The deliveries will stretch over three years.^{[43][44]} By December 2021, the Russian Aerospace Forces received six new aircraft of the third order.^{[45][46][47]}

Su-34M (2022–present)

In June 2022, the Russian Aerospace Forces received another four aircraft of the third order. These were the first aircraft built under the new "Su-34M" standard. They became part of the 277th Bomber Aviation Regiment.^{[48][49]}

In August 2022, the Russian Ministry of Defence signed a new contract for supply of Su-34 bombers.^{[50][51]}

In November 2022, Russia received a new batch of Sukhoi Su-34s of the modernized Su-34M variant; the number is yet unknown but it is safe to assume it was at least four units.^{[52][53][54]} Another batch of modernized Su-34Ms was handed over to the Russian Aerospace Forces in late December 2022. At this point, the total number of Su-34s produced under all contracts, including seven prototypes and pre-production units, reached at least 153 units.^[1] New batches were delivered in June, October and November 2023 and in April, June, September, October, November and December 2024 with at least 10 aircraft delivered in total throughout the year.^{[55][56][57][58][59][60][61][62][63]} The first batch for the 2025 was delivered on 18 April,^[64] the second in July,^{[65][66]} the third in August,^[67] the fourth in September,^[68] the fifth in October,^[69] the sixth in November,^[70] and the seventh in December.^[71]

Design

The Su-34 shares most of its wing structure, tail, and engine nacelles with the Su-27/Su-30, with canards like the Su-30MKI, Su-33 and Su-27M/35 to increase static instability (higher manoeuvrability) and to reduce trim drag.

The Su-34 is powered by a pair of Saturn AL-31FM1 turbofan engines, the same engines used on the Su-27SM, giving the aircraft a maximum speed of Mach 1.8+ when fully loaded.^[72] Although slower than the standard Su-27, the Su-34 can still handle high G-loads and perform aerobatic maneuvers.^[13] When equipped with a full weapons load, the Su-34 has a maximum range of 4,000 km (2,500 mi; 2,200 nmi), or further with aerial refueling.^[73] The airframe is cleared to perform maneuvers of up to +9 g.^[74] The noise level of the Su-34 is two times lower than the level of its predecessors.^[75]



A Russian Air Force Su-34

The Su-34 is a three lifting surface design having both a conventional horizontal tailplane at the rear and a canard foreplane in front of the main wings. The foreplane provides both additional lift (force) and greater maneuverability. It has twin tail fins like those of Su-27 from which it is derived. The Su-34 has 12 hardpoints for 12,000–14,000 kg (26,000–31,000 lb) of ordnance,^[76] intended to include the latest Russian precision-guided weapons. It retains the 30 mm GSh-30-1 cannon of the Su-27 and Su-30,^[77] and the ability to carry six R-77 or R-73 air-to-air missiles, with these being primarily for defense against other fighters. The maximum weight of any single munition carried is 4,000 kg (8,800 lb). Its stand-off weapons have a range of up to 250 km (160 mi; 130 nmi).^{[78][79]} A Khibiny electronic countermeasures (ECM) system is fitted as standard.^[80]



Tail cone between the two engine exhaust nozzles, which houses an auxiliary power unit (APU)



A Sukhoi Su-34 cockpit simulator

Compared to other members of the Flanker family, the Su-34 has an entirely new "Duckbill" nose and forward fuselage designed to increase cabin room and maximize crew comfort and safety, giving the Su-34 the nickname "Duckling", "Hellduck" or "Platypus".^{[13][81]} The two pilots sit side-by-side in NPP Zvezda K-36DM ejection seats: the pilot-commander on the left with the navigator and weapon operator on the right. An advantage of the side by side cockpit is that duplicate instruments are not required for each pilot. Since long missions require comfort, the pressurization system allows operation up to 10,000 metres (32,800 ft) without oxygen masks, which are available for emergencies and combat situations.^[82]

The crew members have room to stand and move about the cabin during long missions.^{[83][84]} The space between the seats allows them to lie down, if necessary.^[82] A hand-held urinal "toilet" and vacuum flask "kitchen" are provided.^[85] A ladder attached to the nose landing gear and a hatch in the cockpit floor is used to enter the cockpit. The cockpit is a continuous capsule of armour (17 mm^[86]).

The multifunctional Leninets V004 main radar has terrain-following and terrain avoidance modes.^[9] Maximum detection range for the passive electronically scanned array radar is 200–250 km (120–160 mi; 110–130 nmi) against large surface targets.^[9] The main radar can simultaneously track ten air targets^[87] and attack four targets (in the air, on land or on the water).^[73] The radius of detection for fighter-sized targets is up to 120 km, the range of the survey is +/- 60 degrees.^[88] The Su-34 reportedly has a frontal radar cross-section that is an order of magnitude smaller than prior generation fighters.^[73]

A new 4th generation radar Pika-M of the complex BKR-3, having a range up to 300 km, passed state tests in 2016.^[89] As of 2021, several aircraft of the RuAF have passed modernization with special equipment that increases capabilities in detecting air and ground targets and expand the types of weapons used.^{[90][91]}

Operational history

Russia

Trials, early service

The Su-34 is rumoured to have made its combat debut during the 2008 Russo-Georgian War.^[92]

In July 2010, several Su-34s and Su-24Ms conducted a 6,000 km (3,700 mi; 3,200 nmi) non-stop flight from air bases in European Russia to the Russian Far East. The exercise included aircraft carrying weapons at full load and simulated delivering them on targets, demonstrating the Su-34's long-range capability. The Su-24Ms were refuelled three times, while the Su-34s were refuelled twice.^[93]

The final stage of the Su-34's state trials was completed on 19 September 2011 and the aircraft entered service in early 2014.^{[94][95]}

Crashes, other accidents

- On 4 June 2015, an Su-34 slid off the runway and flipped over when its parachute failed to open after landing. Nobody was injured. The aircraft was returning from a routine training mission in Russia's Voronezh region.^[96] In June 2016, the damaged aircraft was transferred on board an Antonov An-124 to the Novosibirsk Aircraft Production Association plant for repairs, and was likely returned to service the same year.^[97]
- On 18 January 2019, two Su-34s collided mid-air in Russia's Far East region, while performing a scheduled training flight. Two pilots were killed, one was rescued and one remained missing.^[98] All Su-34 flights were suspended throughout Russia following the accident.^{[99][100]}
- On 6 September 2019, two Su-34s again collided mid-air, this time in the western Russian city of Lipetsk. According to a Russian source the collision happened due to pilot error. Both pilots managed to land after the accident.^{[101][102]}



Sukhoi Su-34 fighter-bomber

- On 21 October 2020, an Su-34 crashed performing a training flight in Khabarovsk region.^[103]
- On 17 October 2022, a Russian Su-34 fighter-bomber crashed into the courtyard of a residential apartment complex in the Russian city of Yeysk during a training flight, setting two blocks on fire. 15 people were killed and 19 were treated in hospital. A school was evacuated. Russian defense ministry sources stated the cause of the crash was an engine fire on takeoff and that both pilots ejected safely. Locals were seen rescuing one of the pilots. Russian naval aviation uses Yeysk as a main training area.^{[104][105][106][107]}
- On 20 April 2023, an Su-34 of the Russian Aerospace Forces was performing a flight over the Russian city of Belgorod when a release of an air ordnance occurred, shattering windows, damaging cars and leaving a crater of about 20 meters in diameter. Three people were injured.^[108]
- On 20 September 2023, an Su-34 "Red 05", RF-95806 crashed in Russia's Voronezh region, without any damage to personnel or ground structures. Both pilots ejected safely.^{[109][110]}
- On 10 June 2024, an Su-34 crashed during a training flight in the mountains of North Ossetia, killing both pilots.^[111]

Syrian Civil War

(2015–2020)

In September 2015, six Su-34s arrived at Latakia airport in Syria, for attacks against rebel and ISIL forces.^[112] Russian air attacks in Syria started on 30 September, in the Homs region.^[113] On 1 October, the Su-34 was used to bomb Islamic State targets in Syria. The Russian Aerospace Forces Su-34 fighter-bombers destroyed an Islamic State command center and training camp south-west of the city of Raqqa. These included precision strikes from an altitude of over 5,000 m (16,400 ft).

Russian Su-34 and Su-25 attack aircraft carried out air strikes the next day against Islamic State targets in Syria's Hama province using precision bombs. According to Russian Defense Ministry spokesman Maj. Gen. Igor Konashenkov, Su-34s hit an ISIL fortified bunker in the Hama province with guided bombs. Fortifications, ammunition depots, seven units of the military equipment near Syria's Maarrat al-Numan were also destroyed by the Russian Aerospace Forces. An ISIL command center and underground depot were also destroyed with explosives near Raqqa.

Defense Ministry spokesman, Maj. Gen. Igor Konashenkov, said in a statement on 3 October: "Accurate delivery of a concrete-piercing bomb BETAB-500 launched from an Su-34 aircraft near Raqqa destroyed a hardened command centre of one of the illegal armed groups as well as an



An Su-34, bottom view



An Su-34 at Latakia International Airport, Syria, in late 2015, early in Russia's intervention in Syria

underground bunker with explosives and an ammunition depot." A Russian Aerospace Forces representative stated Su-34s acquire targets using the GLONASS satellite system for bombing. During this time six Su-34s were in Syria.^[114]

Following the downing of an Su-24 by Turkey, Russia announced on 30 November 2015 that Su-34s in Syria had begun flying combat missions while armed with air-to-air missiles.^[115] On 16 August 2016, Tu-22M3 long-range bombers and Su-34 bombers, having taken off from their base in Hamadan, Islamic Republic of Iran, carried out group airstrikes against targets belonging to ISIS and Jabhat al-Nusra terrorist groups in the provinces of Aleppo, Deir ez-Zor and Idlib.^[116]

On 3 October 2017, Russian Su-34s and Su-35s were deployed to strike "the place of the Al-Nusra leadership meeting" whose location was discovered by Russian military intelligence in Syria subsequently eliminating 12 Al-Nusra field commanders and some 50 militants of the group including Ahmad al-Ghizai, Al-Nusra's security chief according to the Russian Defense Ministry.^{[117][118]}

On 28 May 2018, it was reported that Russian Su-34s intercepted two Israeli Air Force F-16s over Tripoli, Lebanon, forcing them to retreat.^[119]

On 27 February 2020, two Russian Su-34s allegedly carried out an airstrike on a Turkish military convoy killing up to 36 soldiers and injuring at least 60 more in Balyun, Syria.^[120]

On 6 October 2020, two Russian Su-34s conducted air strikes against underground structures in the Eastern bank of Euphrates, Deir Ez-Zor province allegedly inside Rojava territory resulting in the death of ISIL field commander Abu Qatada, who was involved in preparing an attack against Russian forces which resulted in the death of Major General Vyacheslav Gladkikh. Additionally, the strike also destroyed two underground shelters and killed some 30 militants.^[121]

Russo-Ukrainian War

Russian invasion of Ukraine (2022–present)

On 28 February 2022, footage emerged reportedly showing Su-34s overflying the Kharkiv region of Ukraine.^[122]

Until the deployment of the Russian GPS-guided UMPK (Unified Gliding and Correction Module) bomb kits in 2023, a lack of guided bombs required the Su-34s to fly low for accurate bombing, where they were subjected to heavy Ukrainian air-defenses.^[123]

As of 16 February 2025, there have been 36 Su-34s and 1 Su-34M visually confirmed as being lost, damaged or abandoned by Russian forces since the start of the 2022 Russian invasion of Ukraine (29 destroyed in combat, 4 destroyed in a non-combat related incident and 4 damaged).^[124] Ukraine launched several surprise attacks^[125] on Russian Su-34 fighter bombers in the occupied south of Kherson and Mariupol using American-supplied long-



A Russian Su-34 with bombs shot down over Chernihiv on 5 March 2022

range MIM-104 Patriot missiles, causing significant losses of Russian Su-34. As Ukraine fought on, Russia lost almost 36 Su-34 fighter bombers until June 2024.^{[126][127][128]} According to Oryx (website), 41 Su-34s have been damaged or destroyed as of July 2025.^[129]

Dated list of Su-34s shot down in the Russo-Ukrainian War

2022

- The first Su-34 shot down during combat occurred a few days after the beginning of the war on 28 February 2022, near the Buzova airfield.^{[130][131][132]}
- On 1 March, another Su-34 "Red 31", RF-81251 was lost near Borodyanka, Kyiv region.^{[133][134][135]}
- On 5 March, a Su-34 "Red 26", RF-81864 was lost in the area of Hrabivka, Chernihiv Oblast. Both crew members did not survive.^{[136][137][138]}
- On 6 March, a Su-34 RF-95070 was lost over the city of Kharkiv.^{[139][140]}
- On 10 March, a Su-34 reportedly crashed near Kozhan-Gorodok, Belarus, 17 km from the Luninets (air base). Both pilots ejected safely.^[141]
- On 14 March, a Su-34 "Red 35", RF-95010 was lost somewhere over Ukraine.^{[142][143]}
- On 21 April, a Su-34 "Red 43", RF-95858 was lost in Zaporizhzhia Oblast.^{[144][145]}
- On 25 April, a Su-34 "Red 24", RF-95808 was lost near Balakliya, Kharkiv Oblast.^{[146][147]}
- On 18 May, a Su-34 was lost near Kupiansk, Kharkiv Oblast. Senior lieutenant Fetisov Vladimir Nikolayevich died during the crash.^{[148][149][150]}
- On 17 July, a modernized Su-34M reportedly "Red 51", RF-95890 was shot down due to a friendly fire incident near Alchevsk, Luhansk Oblast.^[151]
- On 10 September, wreckage of a Su-34 "Red 20", RF-95004 was found near Izyum, Kharkiv Oblast. It is not known when the plane was shot down.^{[152][153]}
- On 11 September, a Su-34 crashed in the area of Vorontsivka, Krasnoperekopsky district, Crimea. Both pilots ejected safely.^[154]
- On 24 September, a Su-34 "Red 22", RF-95005 was lost over Petropavlivka, Kharkiv Oblast. Both crew members ejected and survived.^[155]
- On 2 October, wreckage of a Su-34 "Red 09", RF-81852 was discovered in Lyman after Russian retreat. Exact crash date is not known.^[156]

2023

- On 3 March, a Su-34 was shot down over Yenakiieve, Donetsk Oblast. One crew member, major Alexander Bondarev, did not survive.^[157]
- On 13 May, a Su-34 from the 47th Aviation Regiment was shot down over Starodubsky District, Bryansk Oblast, Russia. Both crew members died.^[158]
- On 17 December, a Su-34 parked at Morozovsk air base sustained minor damage due to Ukrainian drone attack. The aircraft was protected by anti-drone netting and was covered by ice build up on its airframe.^[159]
- On 22 December, Ukrainian official sources claimed shot down of three Russian Su-34 in southern Ukraine by surface-to-air missiles. At least one loss was confirmed by Russian Telegram channel Fighterbomber, associated with the Russian Aerospace Forces.^{[160][161]}

2024

- On 4 January, it was reported a saboteur attempted to set fire to a Su-34 parked at the Chelyabinsk Shagol Airport. A video of the act was published by Ukrainian GUR.^[162] The saboteur attempted to ignite a fire inside the right air intake of the aircraft but was unsuccessful due to distance between the air intake itself and the engine. The following day on 5 January 2024, it was reported that the Russian Federal Security Service detained a Russian national who was allegedly involved in the sabotage.^[163]

Unconfirmed losses claimed by Ukraine

- On 30 January 2024, Ukrainian forces claimed a Russian Su-34 was shot down over Luhansk Oblast, Eastern Ukraine.^[164] Mykola Oleshchuk, a Ukrainian General said, "Units of the Air Forces of the Armed Forces of Ukraine destroyed three enemy aircraft at once—two Su-34 fighter-bombers and one Su-35 fighter," on Telegram on February 17. He said Ukraine shot down another Russian Su-34 fighter-bomber on the morning of February 18. Oleshchuk said that Ukrainian forces took down additional Su-34 fighter bombers, the seventh Su-34 plane reportedly destroyed by Ukraine within the week. On 27 February 2024 Oleshchuk reported that Ukrainian forces shot down a Russian Su-34 on the eastern front. Later that day another Russian Su-34 was reportedly shot down by Ukrainian forces.^[165] On 29 February Ukraine claimed it shot down three Su-34s.^{[166][167]} In total in February 2024 the Ukrainian Air Force claimed it had shot down ten Su-34 aircraft.^[166] As of 27 February 2024, none of the claims have been independently verified.^[168]
- On the night of 13–14 June 2024, officials from the Ukrainian GUR claimed that two Su-34s were damaged during a Ukrainian drone strike on the Morozovsk air base. There are reports of casualties however these were not officially confirmed.^[169]
- On 14 August 2024, the Ukrainian military claimed to have shot down a Su-34 over Kursk Oblast.^[170]
- On 27 June 2025, the Ukrainian military claimed to have destroyed four Russian Su-34 aircraft at Marinovka airfield, Volgograd region of Russia.^[171]
- On the night of 24–25 September 2025, the Ukrainian military claimed to have destroyed a Russian Su-34 over the city of Zaporizhzhia.^[172]

Non-military usage

Sukhoi Su-34s were used on bombing runs against ice jams on the rivers in Vologda Oblast to prevent floods during the spring of 2016.^{[173][174][175]}

Algerian purchase

In January 2016, Algeria was negotiating a purchase of 12 aircraft for its Air Force.^[176] The deal should have been signed in late 2016,^[177] but in 2017 the talks were still ongoing and no sale was made.^{[178][179]} On 27 December 2019, Algeria reportedly signed a contract for 14 aircraft as part of large military deal that also includes purchase of Su-35 and Su-57 fighters.^[180] However, neither the Russian nor Algerian governments ever confirmed that such a deal had been made.

According to Mil.press, after eight years of fruitless negotiations with Algeria on selling Su-32 (export version), in 2019 Russia agreed to sell the modernized, non-downgraded Su-34ME version; Algeria became interested in the model after the campaign in Syria, where the aircraft demonstrated excellent combat qualities. Production of the first batch was underway as of March 2021 and, according to CAWAT, should be ready in 2021. Training of Algerian pilots would commence in 2022, with all planes

to be delivered by 2023.^[181] According to a 17 March 2021 news article from Middle East Monitor (MEMO), a representative of the Federal Service for Military-Technical Cooperation (FSVTS) in Russia denied the claim that Russia was manufacturing aircraft for Algeria.^[182]

However, the appearance of a Su-34 in a new desert brown and yellow color scheme in mid-May 2025 has fueled speculation that the aircraft have for the first time begun to be produced for export, most likely to Algeria, but possibly to another client with a similar climate such as Iran or Sudan.^[183]

Variants

- **Su-34** – basic aircraft model.
- **Su-32** – export model offered to Algeria in 2012. Not currently offered.^[184]
- **Su-34E** – current export model^[185]
- **Su-34M / Su-34 NVO** (Навигация и Внешнее Оборудование, *Navigation and External Equipment Upgrade*) – modernized version, also known as *Su-34 Sych* (Russian for *Athene*). The Russian contract was signed in 2020 and replaced basic Su-34 in production. According to reports, this upgrade seriously affects almost all of the aircraft's avionics. The radar, sight, and communication systems improved. The range of guided munitions that the bomber can use significantly expanded.^{[186][187]} EW capabilities also expanded with introduction of new hardpoints.^[188] The version was reportedly armed with long range cruise missiles in September 2023.^[189] An armed reconnaissance version entered service in July 2025.^[190]

Operators

Russia

- Russian Aerospace Forces
 - 47th Bomber Aviation Regiment – Baltimor Air Base, Voronezh Oblast^[191]
 - 277th Bomber Aviation Regiment – Khurba Air Base, Khabarovsk Krai^[192]
 - 559th Bomber Aviation Regiment – Morozovsk Air Base, Rostov Oblast^[193]
 - 2nd Guards Bomber Aviation Regiment – Chelyabinsk Shagol Airport, Chelyabinsk Oblast^[194]
 - 4th Centre for Combat Employment and Retraining of Personnel – Lipetsk Air Base, Lipetsk Oblast^{[195][196]}
 - State Flight Testing Center named after V.P. Chkalov – Akhtubinsk Air Base, Astrakhan Oblast^[197]
 - Khmeimim Air Base, Latakia, Syria^[198]

Algeria

- Algerian Air Force
 - The first Su-34ME aircraft were reportedly delivered in 2025.^[199] Algeria originally ordered 14 Su-34ME fighters from Russia.^[200] In May 2026 a footage released by

Russian sources confirmed for the first time the arrival of Su-34MEs to the Algerian air force.^[201]

Specifications (Su-34)

Data from Sukhoi,^[202] Gordon and Davison,^[203]
airforce-technology.com,^[204] airwar.ru,^[205]
ria.ru,^[206] deagel.com^[207]

General characteristics

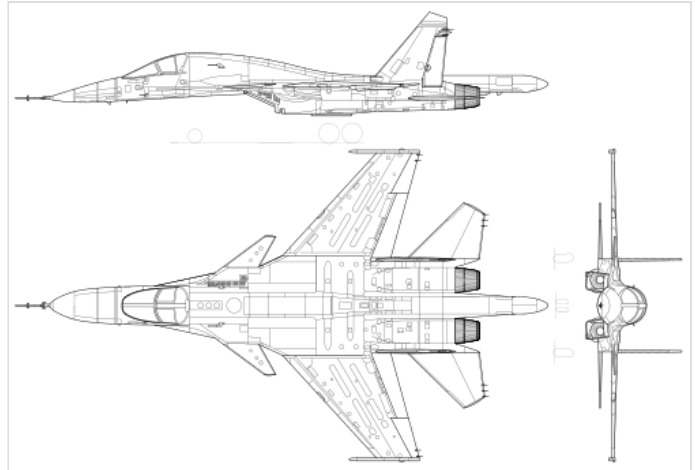
- **Crew:** 2 (side by side seating)
- **Length:** 23.34 m (76 ft 7 in)
- **Wingspan:** 14.7 m (48 ft 3 in)
- **Height:** 6.09 m (20 ft 0 in)
- **Wing area:** 62.04 m² (667.8 sq ft)
- **Airfoil:** 5%^[208]
- **Empty weight:** 22,500 kg (49,604 lb)
- **Gross weight:** 39,000 kg (85,980 lb)
- **Max takeoff weight:** 45,100 kg (99,428 lb)
- **Fuel capacity:** 12,100 kg (26,676 lb) internals
- **Powerplant:** 2 × Saturn AL-31FM1^[209] afterburning turbofan engines, 132 kN (30,000 lbf) with afterburner

Performance

- **Maximum speed:** 1,900 km/h (1,200 mph, 1,000 kn) (Mach 1.8) at altitude
1,435 km/h (892 mph; 775 kn) / Mach 1.2 at sea level^[207]
- **Cruise speed:** 1,300 km/h (810 mph, 700 kn)
- **Combat range:** 1,100 km (680 mi, 590 nmi) (radius of action with standard 12,000 kg weapons load)^{[210][211]}
- **Ferry range:** 4,500 km (2,800 mi, 2,400 nmi)
- **Service ceiling:** 17,000 m (56,000 ft) ^[212]
- **g limits:** +9^[213]
- **Thrust/weight:** 0.68

Armament

- **Guns:** 1 × 30 mm Gryazev-Shipunov GSh-30-1 autocannon with 180 rounds^[214]
- **Hardpoints:** 12 × on wing and fuselage with a capacity of 12,000–14,000 kilograms (26,000–31,000 lb)^{[215][76]}



Su-34 three-view illustration



A front view of the Su-34



Fixing KAB-500S GLONASS Satellite-guided bombs to a Sukhoi Su-34 at Latakia International Airport, 2015

- B-8 rocket pods for 20 × S-8KOM/OM/BM
- B-13 rocket pods for 5 × S-13T/OF
- O-25 rocket pods for 1 × S-25OFM-PU
- S-80FP, with provisions to carry combinations of:
- **Missiles:** Air-to-air missiles:
 - R-27R/ER/T/ET
 - R-73
 - R-77
 - Air-to-surface missiles:
 - Kh-25ML/MT
 - Kh-29L/T/D
 - Kh-31A/AD
 - Kh-35U
 - Kh-38MAE/MKE/MLE/MTE
 - Kh-58
 - Kh-59ME/MK/MK2
 - Kh-47M2^[216]
 - Anti-ship missiles:
 - P-800 Oniks^[217] weight of 1500 kg with a range of up to 300 km and a speed in the range of numbers M= 2.2–3.0. Officially not in service.^{[217][73]}
 - Kh-41
 - Anti-radiation missiles:
 - Kh-25MP
 - Kh-58
 - Kh-31P/PD
 - Cruise missiles:
 - Kh-36 Grom-E1
 - Kh-65S/SE
 - Kh-SD
- **Bombs:** ***KAB-500Kr TV-guided bomb
 - KAB-500L laser-guided bomb
 - KAB-500OD guided bomb
 - KAB-500S-E satellite-guided bomb
 - KAB-1500Kr TV-guided bomb
 - KAB-1500L laser-guided bomb
 - OFAB-250-270 bomb
 - OFAB-100-120 bomb
 - FAB-500T general-purpose bomb
 - FAB-250 UMPK gliding bomb
 - FAB-500 UMPK gliding bomb
 - FAB-1500 UMPK gliding bomb^[218]

- FAB-3000 UMPK gliding bomb^[219]
- 2 × BETAB-500SHP
- P-50T bomb
- ODAB-500PM bomb
- RBK-500 cluster bomb
- SPBE-D bomb
- UMPB D-30SN glide bomb^[220]

Avionics

- V004 passive electronically scanned array radar
- Khibiny electronic countermeasures system
- SAP-14 electronic countermeasures system
- SAP-518 electronic countermeasures system
- UKR-RT SIGINT radio surveillance system
- L150 Pastel radar warning receiver

See also



Related development

- Sukhoi Su-27
- Sukhoi Su-30
- Sukhoi Su-33
- Sukhoi Su-35
- Sukhoi Su-37

Aircraft of comparable role, configuration, and era

- Sukhoi Su-24
- Boeing F/A-18E/F Super Hornet
- McDonnell Douglas F-15E Strike Eagle
- Shenyang J-16
- Xi'an JH-7
- General Dynamics F-111, FB-111

References

1. "ВКС России получили еще три фронтовых бомбардировщика Су-34М" (<https://bmpd.livejournal.com/4637843.html>). *bmpd.livejournal.com*. 1 January 2023. Archived (<https://web.archive.org/web/20230102130934/https://bmpd.livejournal.com/4637843.html>) from the original on 2 January 2023. Retrieved 2 January 2023.

2. "ВКС России получили новые фронтовые бомбардировщики Су-34М" (<https://bmpd.livejournal.com/4707333.html>). 1 June 2023. Archived (<https://web.archive.org/web/20230604085847/https://bmpd.livejournal.com/4707333.html>) from the original on 4 June 2023. Retrieved 4 June 2023.
3. "ВКС России получили вторую в 2024 году партию фронтовых бомбардировщиков Су-34" (<https://bmpd.livejournal.com/4835108.html>). *bmpd* (in Russian). 18 June 2024. Retrieved 20 June 2024.
4. "Su-34 in the sky" (<https://web.archive.org/web/20110929104355/http://www.sukhoi.org/eng/news/company/?id=843>). *Sukhoi*. 2 October 2006. Archived from the original (<http://www.sukhoi.org/eng/news/company/?id=843>) on 29 September 2011. Retrieved 9 January 2011.
5. "Russian Air Force receives 4 new Su-34 fighter-bombers" (<https://web.archive.org/web/20110929104316/http://www.sukhoi.org/eng/news/company/?id=3826>). *Sukhoi*. 24 December 2010. Archived from the original (<http://www.sukhoi.org/eng/news/company/?id=3826>) on 29 September 2011. Retrieved 9 January 2011.
6. "Su-34 frontline bomber was put into service by Russian air forces" (<http://www.ruaviation.com/news/2014/3/20/2236/>). Russian Aviation. 20 March 2014. Archived (<https://web.archive.org/web/20141024062430/http://www.ruaviation.com/news/2014/3/20/2236/>) from the original on 24 October 2014. Retrieved 14 November 2014.
7. "NATO tracks large-scale Russian air activity in Europe" (http://www.nato.int/cps/en/natohq/news_114274.htm). NATO. 29 October 2014. Archived (https://web.archive.org/web/20150910221737/http://www.nato.int/cps/en/natohq/news_114274.htm) from the original on 10 September 2015. Retrieved 24 November 2015.
8. "Su-32" (<https://web.archive.org/web/20110630233848/http://www.sukhoi.org/eng/planes/military/su32/>). *Sukhoi*. Archived from the original (<http://www.sukhoi.org/eng/planes/military/su32/>) on 30 June 2011. Retrieved 4 July 2011.
9. Kopp, Carlo (January 2007). "Sukhoi Su-34 Fullback; Russia's New Heavy Strike Fighter" (<http://www.ausairpower.net/APA-Fullback.html>). *Air Power Australia*: 1. Archived (<https://web.archive.org/web/20150317021534/http://www.ausairpower.net/APA-Fullback.html>) from the original on 17 March 2015. Retrieved 12 December 2018.
10. "Russia's SU-32/34 Long-Range Strike Fighters" (<http://www.defenseindustrydaily.com/russia-to-begin-receiving-su34-longrange-strike-fighters-02595/>). *Defense Industry Daily*. 11 June 2014. Archived (<https://web.archive.org/web/20150118104741/http://www.defenseindustrydaily.com/russia-to-begin-receiving-su34-longrange-strike-fighters-02595/>) from the original on 18 January 2015. Retrieved 18 January 2015.
11. "SU-34 Fullback Supersonic Strike Fighter" (<https://web.archive.org/web/20150118085207/http://defense-update.com/products/s/su-34.htm>). *Defense Update*. 27 October 2006. Archived from the original (<http://defense-update.com/products/s/su-34.htm>) on 18 January 2015. Retrieved 18 January 2015.
12. "Outlook 2010: Reorganization Nears Completion, But Russian Industry Still Has Far To Go" (<http://aviationweek.com/awin/reorganization-nears-completion-russian-industry-still-has-far-go>). *Aviation Week & Space Technology*. Vol. 172, no. 4. 25 January 2010. Archived (<https://web.archive.org/web/20171014134542/http://aviationweek.com/awin/reorganization-nears-completion-russian-industry-still-has-far-go>) from the original on 14 October 2017. Retrieved 24 November 2015.
13. Litovkin, Nikolai (6 June 2017). "Russia's 'Duckbill': Su-34 combines features of both bomber and fighter" (https://www.rbth.com/defence/2017/06/06/russias-duckbill-su-34-combines-features-of-both-bomber-and-fighter_777707). *rbth.com*. Archived (https://web.archive.org/web/20171014084144/https://www.rbth.com/defence/2017/06/06/russias-duckbill-su-34-combines-features-of-both-bomber-and-fighter_777707) from the original on 14 October 2017.

14. Kedrov, Ilya (25 April 2014). "Национальная оборона / Вооружения / Су-34 официально встал в строй ВВС РФ" (<http://www.oborona.ru/includes/periodics/armament/2014/0317/151412845/detail.shtml>) [National defence/Armaments/Su-34 officially stood up operational Russian Air Force]. *oborona.ru* (in Russian). Archived (<https://web.archive.org/web/20150328232116/http://www.oborona.ru/includes/periodics/armament/2014/0317/151412845/detail.shtml>) from the original on 28 March 2015. Retrieved 7 April 2015.
15. Williams (2002), p. 132.
16. Gordon (1999), p. 92.
17. Gordon (1999), p. 93.
18. Eden (2004), p. 466.
19. Andrews, Thomas, "Sukhoi Su-27/Su-30 Family", *International Air Power Review*, Volume 8, Spring 2003.
20. Velovich, Alexander; Barrie, Douglas (22 January 1997). "Radar tests get under way on Sukhoi Su-27IB variant" (<http://www.flightglobal.com/pdfarchive/view/1997/1997%20-%200182.html>). *Flight International*. Archived (<https://web.archive.org/web/20140726195642/http://www.flightglobal.com/pdfarchive/view/1997/1997%20-%200182.html>) from the original on 26 July 2014. Retrieved 9 July 2011.
21. "The defense Ministry is modernizing the bomber su-34" (<https://web.archive.org/web/20160826035402/http://rushincrash.com/forces/the-defense-ministry-is-modernizing-the-bomber-su-34/>). RIA Novosti. 4 July 2016. Archived from the original (<http://rushincrash.com/forces/the-defense-ministry-is-modernizing-the-bomber-su-34/>) on 26 August 2016.
22. "Su-34" (http://www.deagel.com/Combat-Aircraft/Su-34_a000181001.aspx). Deagel.com. Archived (https://web.archive.org/web/20170629064738/http://www.deagel.com/Combat-Aircraft/Su-34_a000181001.aspx) from the original on 29 June 2017. Retrieved 12 December 2018.
23. Sukhankin, Sergey (26 September 2017). "'Syrian Lessons' and Russia's 'Asymmetric Response' to the US" (<https://jamestown.org/program/syrian-lessons-and-russias-asymmetric-response-to-the-us/>). *Jamestown*. Archived (<https://web.archive.org/web/20181021213933/https://jamestown.org/program/syrian-lessons-and-russias-asymmetric-response-to-the-us/>) from the original on 21 October 2018. Retrieved 12 December 2018.
24. Butowski, Piotr (31 October 2018). "Russia Pursues ISR And Electronic Warfare Versions of Su-34 | Defense content from" (<http://aviationweek.com/defense/russia-pursues-isr-and-electronic-warfare-versions-su-34-5>). *Aviation Week*. Archived (<https://web.archive.org/web/20181105052554/http://aviationweek.com/defense/russia-pursues-isr-and-electronic-warfare-versions-su-34-5>) from the original on 5 November 2018. Retrieved 12 December 2018.
25. T. N. I. Staff (4 June 2018). "Meet the Su-34, Russia's Supersonic Strike Aircraft NATO Fears" (<https://nationalinterest.org/blog/the-buzz/meet-the-su-34-russias-supersonic-strike-aircraft-nato-fears-26121>). *The National Interest*. Archived (<https://web.archive.org/web/20181117105333/https://nationalinterest.org/blog/the-buzz/meet-the-su-34-russias-supersonic-strike-aircraft-nato-fears-26121>) from the original on 17 November 2018. Retrieved 4 January 2019.
26. "Historical Background: Su-32" (<https://web.archive.org/web/20151203135136/http://www.sukhoi.org/eng/planes/military/su32/history/>). *Sukhoi*. Archived from the original (<http://www.sukhoi.org/eng/planes/military/su32/history/>) on 3 December 2015. Retrieved 24 November 2015.
27. "The Russian air force received two new Su-34" (<https://web.archive.org/web/20110813203029/http://www.vedomosti.ru/tech/news/2009/12/21/912023>) [Russian Air force received two Su-34]. *Vedomosti* (in Russian). 21 December 2009. Archived from the original (<http://www.vedomosti.ru/tech/news/2009/12/21/912023>) on 13 August 2011.

28. "Russia producing new Su-34 bombers" (http://www.upi.com/Business_News/Security-Industry/2008/01/14/Russia-producing-new-Su-34-bombers/UPI-74801200352690/). *United Press International*. 14 January 2008. Archived (https://web.archive.org/web/20110629134238/http://www.upi.com/Business_News/Security-Industry/2008/01/14/Russia-producing-new-Su-34-bombers/UPI-74801200352690/) from the original on 29 June 2011. Retrieved 7 July 2011.
29. Karnozov, Vladimir. "Russia Orders Another 92 Su-34 Strike Fighters" (<https://www.ainonline.com/aviation-news/defense/2012-03-09/russia-orders-another-92-su-34-strike-fighters>). *Aviation International News*. Archived (<https://web.archive.org/web/20200610124403/https://www.ainonline.com/aviation-news/defense/2012-03-09/russia-orders-another-92-su-34-strike-fighters>) from the original on 10 June 2020. Retrieved 10 June 2020.
30. "Sukhoi completely fulfilled 2012 State Defense Order" (<https://web.archive.org/web/20131112104150/http://www.sukhoi.org/eng/news/company/?id=5042>). *Sukhoi Company (JSC)*. 29 December 2012. Archived from the original (<http://www.sukhoi.org/eng/news/company/?id=5042>) on 12 November 2013. Retrieved 12 November 2013.
31. "Russian Air Force receives first Su-34 bomber" (http://english.pravda.ru/news/russia/06-05-2013/124494-russian_air_force-0/). *English pravda.ru*. 6 May 2013. Archived (https://web.archive.org/web/20131112104042/http://english.pravda.ru/news/russia/06-05-2013/124494-russian_air_force-0/) from the original on 12 November 2013. Retrieved 12 November 2013.
32. "A batch of Su-34 front-line bombers was transferred to the Russian Air Force" (<https://web.archive.org/web/20131112104155/http://sukhoi.org/eng/news/company/?id=5182>). *Sukhoi Company (JSC)*. 9 July 2013. Archived from the original (<http://sukhoi.org/eng/news/company/?id=5182>) on 12 November 2013. Retrieved 12 November 2013.
33. "Sukhoi fulfilled the 5-year state contract with the final batch of Su-34 front-line bombers for the Russian Air Force - RUSSIAN AVIATION" (<https://www.ruaviation.com/news/2013/12/16/2093/?h>). *ruaviation.com*. Archived (<https://web.archive.org/web/20210308194519/https://www.ruaviation.com/news/2013/12/16/2093/?h>) from the original on 8 March 2021. Retrieved 14 March 2022.
34. "Su-34 for the 559th bap (Morozovsk)" (<https://www.youtube.com/watch?v=pa-u2ChSoc0>). *Rossiya 1 via Youtube*. 10 June 2014. Archived (<https://web.archive.org/web/20151204052128/https://www.youtube.com/watch?v=pa-u2ChSoc0>) from the original on 4 December 2015. Retrieved 24 November 2015.
35. "Три новосибирских Су-34 улетели в Ростовскую область – 54новости.рф" (https://web.archive.org/web/20141025034807/http://54novosti.ru/politika_i_ekonomika/tri-novosibirskih-su-34-uleteli-v-rostovskuyu-oblast/) [Three of Novosibirsk Su-34 flew in Rostov Oblast] (in Russian). 18 July 2014. Archived from the original (http://54novosti.ru/politika_i_ekonomika/tri-novosibirskih-su-34-uleteli-v-rostovskuyu-oblast/) on 25 October 2014. Retrieved 14 November 2014.
36. "Новосибирский авиазавод им. Чкалова увеличил план по выпуску бомбардировщиков Су-34" (<http://itar-tass.com/sibir-news/1186096>) [The Novosibirsk plant increased plan for production of Su-34 bombers]. *TASS* (in Russian). 14 March 2014. Archived (<https://web.archive.org/web/20140724065636/http://itar-tass.com/sibir-news/1186096>) from the original on 24 July 2014. Retrieved 7 April 2015.
37. "Sukhoi handed over two Su-34 frontline bombers to the Russian Air Force exceeding initial yearly plan" (<https://web.archive.org/web/20150106222548/http://www.sukhoi.org/eng/news/company/?id=5603>). *Sukhoi Company (JSC)*. 22 December 2014. Archived from the original (<http://www.sukhoi.org/eng/news/company/?id=5603>) on 6 January 2015. Retrieved 7 April 2015.
38. "Su-34 park may reach 200 aircraft" (<http://itar-tass.com/en/russia/712992>). *TASS*. 26 December 2013. Retrieved 7 April 2015.

39. "Sukhoi Delivers More Su-34 Bombers to Russian MoD" (<http://www.forecastinternational.com/news/index.cfm?I3=30101&recno=232627>). *Forecast International*. 21 May 2015. Archived (<https://web.archive.org/web/20150924014609/http://www.forecastinternational.com/news/index.cfm?I3=30101&recno=232627>) from the original on 24 September 2015. Retrieved 24 November 2015.
40. "ПАО "Компания "Сухой" – Новости – Новости компании" (<https://web.archive.org/web/20151004072323/http://www.sukhoi.org/news/company/?id=5721>) [Sukhoi has transferred a batch of Russian air force Su-34]. *Sukhoi* (in Russian). 16 July 2015. Archived from the original (<http://www.sukhoi.org/news/company/?id=5721>) on 4 October 2015. Retrieved 24 November 2015.
41. "Подробности :: Ведомости" (<https://web.archive.org/web/20160214122804/http://vedomosti.sfo.ru/articles/?article=46910>). Archived from the original (<http://vedomosti.sfo.ru/articles/?article=46910>) on 14 February 2016. Retrieved 29 December 2015.
42. "ВКС России получили еще два самолета Су-34" (<https://bmpd.livejournal.com/4203875.html>). *bmpd.livejournal.com*. 5 December 2020. Archived (<https://web.archive.org/web/20220922204143/https://bmpd.livejournal.com/4203875.html>) from the original on 22 September 2022. Retrieved 5 July 2022.
43. "Russian Ministry of Defense signed contracts with UAC at Army-2020 for new aircrafts [sic]" (<https://www.airrecognition.com/index.php/news/defense-aviation-news/2020/august/6530-russian-ministry-of-defense-signed-contracts-with-uac-at-army-2020-for-new-aircrafts.html>). *airrecognition.com*. 28 August 2020. Archived (<https://web.archive.org/web/20210918072006/https://www.airrecognition.com/index.php/news/defense-aviation-news/2020/august/6530-russian-ministry-of-defense-signed-contracts-with-uac-at-army-2020-for-new-aircrafts.html>) from the original on 18 September 2021. Retrieved 31 August 2020.
44. "Минобороны России и Ростех подписали на "Армии-2020" соглашения на сумму более 300 млрд. рублей" (<https://armstrade.org/includes/periodics/news/2020/0826/094559332/detail.shtml>). *armstrade.org*. 26 August 2020. Archived (<https://web.archive.org/web/20230405130314/https://armstrade.org/includes/periodics/news/2020/0826/094559332/detail.shtml>) from the original on 5 April 2023. Retrieved 31 August 2020.
45. "Поставки боевых самолетов в Вооруженные Силы России в 2021 году" (<https://bmpd.livejournal.com/4476065.html>). *bmpd.livejournal.com*. 28 January 2022. Archived (<https://web.archive.org/web/20220705201546/https://bmpd.livejournal.com/4476065.html>) from the original on 5 July 2022. Retrieved 5 July 2022.
46. "В ВВО поступили новые бомбардировщики Су-34" (<https://web.archive.org/web/20211209160640/https://armstrade.org/includes/periodics/news/2021/1207/095565577/detail.shtml>) [Air Defense Forces received new Su-34 bombers]. *armstrade.org* (in Russian). 7 December 2021. Archived from the original (<https://armstrade.org/includes/periodics/news/2021/1207/095565577/detail.shtml>) on 9 December 2021.
47. "В Липецкий авиацентр ВКС поступила партия истребителей-бомбардировщиков Су-34" (<https://armstrade.org/includes/periodics/news/2021/1222/141065808/detail.shtml>) [A batch of Su-34 fighter-bombers arrived at the Lipetsk Aviation Center of the Aerospace Forces]. *armstrade.org* (in Russian). 22 December 2021. Archived (<https://web.archive.org/web/20211223165433/https://armstrade.org/includes/periodics/news/2021/1222/141065808/detail.shtml>) from the original on 23 December 2021. Retrieved 23 December 2021.
48. "ВКС России получили первые в 2022 году фронтовые бомбардировщики Су-34М" (<https://bmpd.livejournal.com/4550723.html>). *bmpd.livejournal.com*. 4 July 2022. Archived (<https://web.archive.org/web/20220705200043/https://bmpd.livejournal.com/4550723.html>) from the original on 5 July 2022. Retrieved 5 July 2022.

49. "Russian Air Force receives first Su-34M front-line bombers" (<https://www.airrecognition.com/index.php/news/defense-aviation-news/2022-news-aviation-aerospace/july/8448-russian-air-force-receives-first-su-34m-front-line-bombers.html>). *airrecognition.com*. 5 July 2022. Archived (<https://web.archive.org/web/20220929132334/https://www.airrecognition.com/index.php/news/defense-aviation-news/2022-news-aviation-aerospace/july/8448-russian-air-force-receives-first-su-34m-front-line-bombers.html>) from the original on 29 September 2022. Retrieved 5 July 2022.
50. "Russia's top brass inks deal on delivery of Su-34 frontline bombers" (<https://tass.com/defense/1494401>). TASS. 16 August 2022. Archived (<https://web.archive.org/web/20220818113354/https://tass.com/defense/1494401>) from the original on 18 August 2022. Retrieved 18 August 2022.
51. "Подписан контракт на поставку партии Су-34" (<https://www.uacrussia.ru/ru/press-center/news/podpisan-kontrakt-na-postavku-partii-su-34>). *uacrussia.ru*. 16 August 2022. Archived (<https://web.archive.org/web/20230102130934/https://www.uacrussia.ru/ru/press-center/news/podpisan-kontrakt-na-postavku-partii-su-34>) from the original on 2 January 2023. Retrieved 2 January 2023.
52. "Defense firm delivers new batch of Su-34 bombers to Russian troops" (<https://tass.com/defense/1537767>). Archived (<https://web.archive.org/web/20221118110113/https://tass.com/defense/1537767>) from the original on 18 November 2022. Retrieved 18 November 2022.
53. "Russian Air Force Receive a New Batch of 'Deadly' Su-34 Fighter Bombers Amid 'Escalating War' with Ukraine" (<https://eurasianimes.com/russian-air-force-receive-a-new-batch-of-deadly-su-34-fighter-bombers/>). 17 November 2022. Archived (<https://web.archive.org/web/20221127031457/https://eurasianimes.com/russian-air-force-receive-a-new-batch-of-deadly-su-34-fighter-bombers/>) from the original on 27 November 2022. Retrieved 27 November 2022.
54. "Russian military receives new Su-34 fighter-bombers" (<https://mil.in.ua/en/news/russian-military-receives-new-su-34-fighter-bombers/>). Archived (<https://web.archive.org/web/20221127031457/https://mil.in.ua/en/news/russian-military-receives-new-su-34-fighter-bombers/>) from the original on 27 November 2022. Retrieved 27 November 2022.
55. "Defense contractor delivers new batch of Su-34 frontline bombers to Russian troops" (<https://tass.com/defense/1626261>). Archived (<https://web.archive.org/web/20230604084723/https://tass.com/defense/1626261>) from the original on 4 June 2023. Retrieved 4 June 2023.
56. "Defense contractor delivers 2nd batch of Su-34 frontline bombers to Russian troops" (<https://tass.com/defense/1686869>). Archived (<https://web.archive.org/web/20231010053018/https://tass.com/defense/1686869>) from the original on 10 October 2023. Retrieved 10 October 2023.
57. "ЦАМТО // ОАК передала ВКС России очередную партию фронтовых бомбардировщиков Су-34" (<https://armstrade.org/includes/periodics/news/2023/1122/095076535/detail.shtml>). *ЦАМТО / Центр анализа мировой торговли оружием* (in Russian). 22 November 2023. Archived (<https://web.archive.org/web/20231123153906/https://armstrade.org/includes/periodics/news/2023/1122/095076535/detail.shtml>) from the original on 23 November 2023. Retrieved 23 November 2023.
58. "Russian forces get new batch of Su-34 fighter-bombers" (<https://defence-blog.com/russian-forces-get-new-batch-of-su-34-fighter-bombers/>). *Defence Blog*. 7 April 2024. Retrieved 7 April 2024.
59. "Russian military receives new Su-34s, but losses remain high" (<https://essanews.com/russian-military-receives-new-su-34s-but-losses-remain-high,7039563821983361a>). *essanews.com*. 17 June 2024. Retrieved 18 June 2024.
60. "ОАК передала очередную партию самолетов Су-34 в войска" (<https://rostec.ru/news/oa-k-peredala-ocherednuyu-partiyu-samoletov-su-34-v-voyska/>). *Ростех* (in Russian). Retrieved 3 September 2024.

61. Nikolov, Boyko (9 October 2024). "Fresh batch of Su-34 bombers ready for action in Russian VKS" (<https://bulgarianmilitary.com/2024/10/09/fresh-batch-of-su-34-bombers-ready-for-action-in-russian-vks/>). *Bulgarian Military Industry Review*. Retrieved 10 October 2024.
62. Nikolov, Boyko (25 November 2024). "Russia unleashes fresh Su-34 fighter-bombers on Ukraine" (<https://bulgarianmilitary.com/2024/11/25/russia-unleashes-fresh-su-34-fighter-bombers-on-ukraine/>). *Bulgarian Military Industry Review*. Retrieved 27 November 2024.
63. "Russian Aerospace Forces Receive More Su-57 Jets in 2024 Vs. 2023" (https://www.defensemirror.com/news/38458/Russian_Aerospace_Forces_Receive_More_Su_57_Jets_in_2024_Vs_2023). *Defense Aerospace news*. 23 December 2024. Retrieved 24 December 2024.
64. Pryhodko, Roman (19 April 2025). "First in 2025: Rostec delivers a batch of Su-34 to the Russian Aerospace Forces" (<https://militaryni.com/en/news/first-in-2025-rostec-delivers-a-batch-of-su-34-to-the-russian-aerospace-forces/>). *Militaryni*. Retrieved 21 April 2025.
65. Defesa, Redação Forças de (15 July 2025). "Rússia recebe novos Sukhoi Su-34 com capacidade para bombas guiadas FAB-UMPK" (https://www.aereo.jor.br/2025/07/15/russia-recebe-novos-sukhoi-su-34-com-capacidade-para-bombas-guiadas-fab-umpk/#goog_rewarded). *Poder Aéreo – Aviação, Forças Aéreas, Indústria Aeroespacial e de Defesa* (in Portuguese). Retrieved 18 July 2025.
66. "Video: Russia Showcases Power with New Su-34 Fighter Jets" (<https://www.msn.com/en-ie/money/technology/video-russia-showcases-power-with-new-su-34-fighter-jets/ar-AA1lluRd>). *MSN*. 18 July 2025. Retrieved 18 July 2025.
67. "Rostec delivers new batch of Su-34 fighters to Russian Army" (<https://defensehere.com/en/rostec-delivers-new-batch-of-su-34-fighters-to-russian-army/>). *Defensehere*. 12 August 2025. Retrieved 13 August 2025.
68. Malyasov, Dylan (15 September 2025). "New Su-34 fighter-bombers delivered to Russian military" (<https://defence-blog.com/new-su-34-fighter-bombers-delivered-to-russian-military/>). *Defence Blog*. Retrieved 16 September 2025.
69. Pryhodko, Roman (7 October 2025). "UAC delivered the fifth batch of Su-34 bombers to the Russian Aerospace Forces" (<https://militaryni.com/en/news/uac-delivered-the-fifth-batch-of-su-34-bombers-to-the-russian-aerospace-forces/>). *Militaryni*. Retrieved 9 October 2025.
70. Kozatskyi, Sania (6 November 2025). "Russian Aerospace Forces Receive Sixth Batch of New Su-34 Bombers" (<https://militaryni.com/en/news/russian-aerospace-forces-receive-sixth-batch-of-new-su-34-bombers/>). *Militaryni*. Retrieved 10 November 2025.
71. Kozatskyi, Sania (10 December 2025). "Russia Receives Seventh Batch of New Su-34 Fighter Jets in 2025" (<https://militaryni.com/en/news/russia-receives-seventh-batch-of-new-su-34-fighter-jets-in-2025/>). *Militaryni*. Retrieved 11 December 2025.
72. "АЛ-31Ф серии 42 (М1)" (<http://www.salut.ru/ViewTopic.php?Id=652>) [AL-31F series 42 (M1)]. *Salut.ru* (in Russian). Archived (<https://web.archive.org/web/20140709152845/http://www.salut.ru/ViewTopic.php?Id=652>) from the original on 9 July 2014. Retrieved 1 March 2012.
73. "Сухой Су-34" (<http://www.airwar.ru/enc/bomber/su34.html>) [Sukhoi Su-34]. *airwar.ru* (in Russian). Archived (<https://web.archive.org/web/20150119183727/http://airwar.ru/enc/bomber/su34.html>) from the original on 19 January 2015. Retrieved 7 April 2015.
74. "Первая в 2014 году партия Су-34 отправилась в войска" (http://www.arms-expo.ru/news/production/pervaya_v_2014_godu_partiya_su_34_otpravilas_v_voyska/) [First in 2014 batch of Su-34 embarked troops]. *arms-expo.ru* (in Russian). 11 June 2014. Archived (https://web.archive.org/web/20150413092843/http://www.arms-expo.ru/news/production/pervaya_v_2014_godu_partiya_su_34_otpravilas_v_voyska/) from the original on 13 April 2015. Retrieved 7 April 2015.

75. "Central Military District Aviation receives two Su-34 multi-role fighters under State Defence Order : Ministry of Defence of the Russian Federation" (http://eng.mil.ru/en/news_page/country/more.htm?id=12200393@egNews). Eng.mil.ru. 18 October 2018. Archived (https://web.archive.org/web/20181020140851/http://eng.mil.ru/en/news_page/country/more.htm?id=12200393@egNews) from the original on 20 October 2018. Retrieved 12 December 2018.
76. "Russian ground forces have been massing around Ukraine for months. Now Russia's air force is joining them" (<https://www.businessinsider.com/russias-air-force-is-joining-military-forces-massing-around-ukraine-2022-2>). *Business Insider*. Archived (<https://web.archive.org/web/20220605200608/https://www.businessinsider.com/russias-air-force-is-joining-military-forces-massing-around-ukraine-2022-2>) from the original on 5 June 2022. Retrieved 5 June 2022.
77. Gordon & Davison (2006), pp. 80–81.
78. "Su-34/Su-32FN Bomber" (<https://web.archive.org/web/20150930233441/http://www.airforceworld.com/bomber/eng/su34.htm>). *airforceworld.com*. Archived from the original (<http://www.airforceworld.com/bomber/eng/su34.htm>) on 30 September 2015. Retrieved 7 April 2015.
79. "ФРОНТОВОЙ БОМБАРДИРОВЩИК СУ-34, FRONT-LINE BOMBER SU-34" (http://bastion-karpenko.narod.ru/Su-32_48.html). *narod.ru* (in Russian). Archived (https://web.archive.org/web/20150315051832/http://bastion-karpenko.narod.ru/Su-32_48.html) from the original on 15 March 2015. Retrieved 7 April 2015.
80. Zettlemoyer, Ryan. "Khibiny Electronic Countermeasures System; Russia's Secret Technology!" (<https://english.iswnnews.com/34871/khibiny-electronic-countermeasures-system-russias-secret-technology/>). *Islamic World News*. Islamic World News. Retrieved 19 May 2025.
81. Newdick, Thomas (21 January 2022). "All The Crazy Quirks And Features On Russia's Su-34 Fullback Strike Fighter" (<https://www.thedrive.com/the-war-zone/43921/all-crazy-quirks-and-features-on-russias-su-34-fullback-strike-fighter>). *The Drive*. Archived (<https://web.archive.org/web/20220304221319/https://www.thedrive.com/the-war-zone/43921/all-crazy-quirks-and-features-on-russias-su-34-fullback-strike-fighter>) from the original on 4 March 2022. Retrieved 8 March 2022.
82. Gordon (1999), p. .
83. Spick (2000), pp. 518–519.
84. Eden (2004).
85. "ФРОНТОВОЙ БОМБАРДИРОВЩИК СУ-34" (<https://web.archive.org/web/20161102233500/http://bastion-karpenko.ru/su-34/>). *bastion-karpenko.ru*. Archived from the original (<http://bastion-karpenko.ru/su-34/>) on 2 November 2016. Retrieved 12 October 2021.
86. "Характеристики бомбардировщика Су-34" (<http://ria.ru/infografika/20150916/1253438900.html>) [Features bomber Su-34]. *RIA Novosty* (in Russian). 16 September 2015. Archived (<https://web.archive.org/web/20151121022801/http://ria.ru/infografika/20150916/1253438900.html>) from the original on 21 November 2015. Retrieved 24 November 2015.
87. Cenciotti, David (4 October 2015). "This cool Infographic says all you need to know about the Sukhoi Su-34 the most advanced Russian plane in Syria" (<https://theaviationist.com/2015/10/04/this-cool-infographic-says-all-you-need-to-know-about-the-sukhoi-su-34-the-most-advanced-russian-plane-in-syria/>). *Theaviationist.com*. Archived (<https://web.archive.org/web/20180927052239/http://theaviationist.com/2015/10/04/this-cool-infographic-says-all-you-need-to-know-about-the-sukhoi-su-34-the-most-advanced-russian-plane-in-syria/>) from the original on 27 September 2018. Retrieved 12 December 2018.
88. "Самолеты" (<http://www.sukhoi.org/products/samolety/254/>). *sukhoi.org*. Archived (<https://web.archive.org/web/20180321130830/http://www.sukhoi.org/products/samolety/254/>) from the original on 21 March 2018. Retrieved 21 March 2018.

89. " "ОПК" запускает серийное производство "всевидающих" РЛС | Еженедельник "Военно-промышленный курьер" " (<http://vpk-news.ru/news/32554>). *Vpk-news.ru* (in Russian). Archived (<https://web.archive.org/web/20170213001638/http://vpk-news.ru/news/32554>) from the original on 13 February 2017. Retrieved 1 July 2017.
90. "ЦАМТО / / Авиаполк ЦВО в Челябинской области пополнился модернизированным истребителем-бомбардировщиком Су-34" (<https://armstrade.org/includes/periodics/news/2021/0312/095561974/detail.shtml>). *armstrade.org*. Archived (<https://web.archive.org/web/20220119053654/https://armstrade.org/includes/periodics/news/2021/0312/095561974/detail.shtml>) from the original on 19 January 2022. Retrieved 14 March 2021.
91. "ЦАМТО / / Боевые возможности российских строевых фронтовых бомбардировщиков Су-34 расширили" (<https://armstrade.org/includes/periodics/news/2021/0412/101062398/detail.shtml>). *armstrade.org*. Archived (<https://web.archive.org/web/20230415090239/https://armstrade.org/includes/periodics/news/2021/0412/101062398/detail.shtml>) from the original on 15 April 2023. Retrieved 15 April 2021.
92. "Пятидневная война: итог в воздухе" (<http://army.lv/?s=2635&id=4636>) [Five-Day War: Outcome in the air]. *Army.lv* (in Russian). Archived (<https://web.archive.org/web/20120105110910/http://army.lv/?s=2635&id=4636>) from the original on 5 January 2012. Retrieved 1 March 2012.
93. "Warplanes: F-15Eski Goes Long Against China" (<http://www.strategypage.com/htmww/htairfo/articles/20100706.aspx>). *Strategypage.com*. 6 July 2010. Archived (<https://web.archive.org/web/20120320025602/http://www.strategypage.com/htmww/htairfo/articles/20100706.aspx>) from the original on 20 March 2012. Retrieved 1 March 2012.
94. Krelani, Vegim (25 September 2022). "Sukhoi Su-34 Fullback: Russia's Agile Bomber" (<http://planehistoria.com/sukhoi-su-34/>). *PlaneHistoria*. Retrieved 24 April 2025.
95. Balestrieri, Steve (30 March 2025). "Russia's Su-34 'Fullback' Fighter-Bomber Hybrid Is In Deep Trouble in Ukraine" (<https://www.19fortyfive.com/2025/03/russia-su-34-fullback-fighter-bomber-hybrid-is-in-deep-trouble-in-ukraine/>). *19FortyFive*. Retrieved 24 April 2025.
96. "Su-34 bomber crashes in Voronezh region; pilots survive" (http://rbth.com/news/2015/06/04/su-34_bomber_crashes_in_voronezh_region_pilots_survive_46646.html). Russia beyond the headlines. 4 June 2015. Archived (https://web.archive.org/web/20171014234150/https://www.rbth.com/news/2015/06/04/su-34_bomber_crashes_in_voronezh_region_pilots_survive_46646.html) from the original on 14 October 2017. Retrieved 25 November 2015.
97. "Перевернувшийся в 2015 году Су-34 доставлен "Русланом" на ремонт в Новосибирск – bmpd" (<http://bmpd.livejournal.com/1964139.html>). *bmpd.livejournal.com*. 16 June 2016. Archived (<https://web.archive.org/web/20160820013337/http://bmpd.livejournal.com/1964139.html>) from the original on 20 August 2016. Retrieved 1 July 2017.
98. "Body of third Su-34 pilot found by rescuers — Russian Defense Ministry" (<https://tass.com/emergencies/1040770>). TASS. Archived (<https://web.archive.org/web/20190728072252/http://tass.com/emergencies/1040770>) from the original on 28 July 2019. Retrieved 28 July 2019.
99. "Two Su-34 jets collide in Russia's Far East" (<http://tass.com/emergencies/1040603>). TASS. 18 January 2019. Archived (<https://web.archive.org/web/20190119121048/http://tass.com/emergencies/1040603>) from the original on 19 January 2019.
100. "All Su-34 flights suspended in Russia following accident in Far East" (<http://tass.com/defense/1040615>). TASS. 18 January 2019. Archived (<https://web.archive.org/web/20190119174216/http://tass.com/defense/1040615>) from the original on 19 January 2019.
101. "Two Russian Su-34 Fullbacks Collided Twice during the Same Training Sortie. Both Aircraft Landed Safely" (<https://theaviationgeekclub.com/two-russian-su-34-fullbacks-collided-twice-during-the-same-training-sortie-both-aircraft-landed-safely/>). 6 September 2019. Archived (<https://web.archive.org/web/20200222113453/https://theaviationgeekclub.com/two-russian-su-34-fullbacks-collided-twice-during-the-same-training-sortie-both-aircraft-landed-safely/>) from the original on 22 February 2020. Retrieved 22 February 2020.

102. "Russia: 2 Su-34 fighter bombers collide" (<https://www.aa.com.tr/en/world/russia-2-su-34-fighter-bombers-collide-/1574727>). *aa.com.tr*. Archived (<https://web.archive.org/web/20200222113445/https://www.aa.com.tr/en/world/russia-2-su-34-fighter-bombers-collide-/1574727>) from the original on 22 February 2020. Retrieved 22 February 2020.
103. "Lostarmour ID: 26187" (<https://lostarmour.info/aflosses/item.php?id=26187>). Archived (<https://web.archive.org/web/20201101040937/https://lostarmour.info/aflosses/item.php?id=26187>) from the original on 1 November 2020. Retrieved 28 October 2020.
104. "Russian warplane crash kills 13 in apartment block" (<https://www.bbc.com/news/world-europe-63294701>). BBC. 18 October 2022. Archived (<https://web.archive.org/web/20221019011200/https://www.bbc.com/news/world-europe-63294701>) from the original on 19 October 2022. Retrieved 18 October 2022.
105. "How did jet crash into a block of flats in Russia?" (<https://www.bbc.com/news/63313871>). *BBC News*. 20 October 2022. Archived (<https://web.archive.org/web/20221021152428/https://www.bbc.com/news/63313871>) from the original on 21 October 2022. Retrieved 21 October 2022.
106. "Aircraft crashes into residential building in Russian city of Yeysk - agencies" (<https://www.devdiscourse.com/article/law-order/2217370-aircraft-crashes-into-residential-building-in-russian-city-of-yeysk---agencies/>). *Devis Discourse / Reuters*. 17 October 2022. Archived (<https://web.archive.org/web/20221017162337/https://www.devdiscourse.com/article/law-order/2217370-aircraft-crashes-into-residential-building-in-russian-city-of-yeysk---agencies/>) from the original on 17 October 2022. Retrieved 17 October 2022.
107. "Russia has lost more than 10% of its Su-34 Fullback fleet in Ukraine ops" (<https://www.key.aero/article/russia-has-lost-more-10-its-su-34-fullback-fleet-ukraine-ops>). *key.aero*. 11 October 2022. Archived (<https://web.archive.org/web/20221026132256/https://www.key.aero/article/russia-has-lost-more-10-its-su-34-fullback-fleet-ukraine-ops>) from the original on 26 October 2022. Retrieved 26 October 2022.
108. "Emergency release of air ordnance occurred during Su-34 flight above Belgoroddate" (<https://tass.com/emergencies/1607497>) Archived (<https://web.archive.org/web/20230421104947/https://tass.com/emergencies/1607497>) 21 April 2023 at the Wayback Machine. *tass.com*
109. "Russia's \$50m Su-34 Fighter Jet Crashes in Voronezh" (<https://www.newsweek.com/russia-fighter-jet-sukhoi-su-34-crash-voronezh-1828386>). *Newsweek*. 20 September 2023. Archived (<https://web.archive.org/web/20230920212134/https://www.newsweek.com/russia-fighter-jet-sukhoi-su-34-crash-voronezh-1828386>) from the original on 20 September 2023. Retrieved 20 September 2023.
110. "Incident Sukhoi Su-34 RF-95806 05 R" (<https://aviation-safety.net/wikibase/345752>). Archived (<https://web.archive.org/web/20240103045809/https://aviation-safety.net/wikibase/345752>) from the original on 3 January 2024. Retrieved 3 March 2024.
111. Barber, Harriet (11 June 2024). "Russian fighter-bomber crashes after 'technical fault' " (<https://www.telegraph.co.uk/world-news/2024/06/11/russian-fighter-bomber-crashes-after-technical-fault/>). *The Telegraph*. ISSN 0307-1235 (<https://search.worldcat.org/issn/0307-1235>). Retrieved 11 June 2024.
112. Cenciotti, David (29 September 2015). "Six Russian Su-34 Fullback bomber have just arrived in Syria. And this is the route they have likely flown to get there" (<http://theaviationist.com/2015/09/29/su-34-have-arrived-in-syria>). *The Aviationist*. Archived (<https://web.archive.org/web/20151129104600/http://theaviationist.com/2015/09/29/su-34-have-arrived-in-syria/>) from the original on 29 November 2015. Retrieved 24 November 2015.
113. Cooper, Helene; Gordon, Michael R.; MacFarquhar, Neil (30 September 2015). "Russians Strike Targets in Syria, but Not ISIS Areas" (<https://www.nytimes.com/2015/10/01/world/europe/putin-military-syria.html>). *The New York Times*. Archived (<https://web.archive.org/web/20240312192534/https://www.nytimes.com/2015/10/01/world/europe/russia-airstrikes-syria.html>) from the original on 12 March 2024. Retrieved 25 November 2015.

114. "Russian weapons in Syria are overrated" (<http://www.dailystar.com.lb/Opinion/Commentary/2015/Oct-23/319914-russian-weapons-in-syria-are-overrated.ashx>). *TheDailyStar.com*. 23 October 2015. Archived (<https://web.archive.org/web/20151123082620/http://www.dailystar.com.lb/Opinion/Commentary/2015/Oct-23/319914-russian-weapons-in-syria-are-overrated.ashx>) from the original on 23 November 2015. Retrieved 25 November 2015.
115. "Russia arms Syrian-based fighters with air-to-air missiles" (<https://www.flightglobal.com/news/articles/russia-arms-syrian-based-fighters-with-air-to-air-mi-419601/>). *FlightGlobal*. Archived (<https://web.archive.org/web/20151208071945/https://www.flightglobal.com/news/articles/russia-arms-syrian-based-fighters-with-air-to-air-mi-419601/>) from the original on 8 December 2015. Retrieved 1 July 2017.
116. "TASS: Military & Defense – Russian bombers first time use Iranian air base for strikes in Syria" (<http://tass.ru/en/defense/894527>). *Tass.ru*. Archived (<https://web.archive.org/web/20160818222538/http://tass.ru/en/defense/894527>) from the original on 18 August 2016. Retrieved 1 July 2017.
117. "Chief of the Main Operational Directorate of the Russian General Staff Colonel General Sergei Rudskoy held a briefing for media representatives" (<https://web.archive.org/web/20220117160052/https://syria.mil.ru/en/index/syria/news/more.htm?id=12146035@egNews>). *mil.ru*. 28 December 2019. Archived from the original (<http://syria.mil.ru/en/index/syria/news/more.htm?id=12146035@egNews>) on 17 January 2022. Retrieved 11 October 2020.
118. "Russian Airstrike in Syria Wounds Jabhat al-Nusra Commander, Kills 65 Terrorists" (https://www.defenseworld.net/news/20845/Russian_Airstrike_in_Syria_Wounds_Jabhat_al_Nusra_Commander_Kills_65_Terrorists). *defenseworld.net*. 25 April 2020. Archived (https://web.archive.org/web/20200919221252/https://www.defenseworld.net/news/20845/Russian_Airstrike_in_Syria_Wounds_Jabhat_al_Nusra_Commander_Kills_65_Terrorists) from the original on 19 September 2020. Retrieved 4 December 2020.
119. "In rare face-off, Russian jets reportedly intercept Israeli planes over Lebanon" (<https://www.timesofisrael.com/russian-jets-reportedly-intercept-israeli-planes-over-lebanon/>). *The Times of Israel*. 28 May 2018. Archived (<https://web.archive.org/web/20190928102326/https://www.timesofisrael.com/russian-jets-reportedly-intercept-israeli-planes-over-lebanon/>) from the original on 28 September 2019. Retrieved 10 September 2019.
120. Stevis-Gridneff, Matina; Gall, Carlotta (29 February 2020). "Erdogan Says, 'We Opened the Doors,' and Clashes Erupt as Migrants Head for Europe" (<https://www.nytimes.com/2020/02/29/world/europe/turkey-migrants-eu.html>). *The New York Times*. Archived (<https://web.archive.org/web/20200302090622/https://www.nytimes.com/2020/02/29/world/europe/turkey-migrants-eu.html>) from the original on 2 March 2020. Retrieved 29 February 2020.
121. "Russia avenges the death of General Vyacheslav Gladkikh in Syria with an air strike" (<https://thetimeshub.in/russia-avenges-the-death-of-general-vyacheslav-gladkikh-in-syria-with-an-air-strike/8098/>). Natasha Kumar, *thetimeshub.in*. 9 October 2020. Archived (<https://web.archive.org/web/20201026101016/https://thetimeshub.in/russia-avenges-the-death-of-general-vyacheslav-gladkikh-in-syria-with-an-air-strike/8098/>) from the original on 26 October 2020. Retrieved 9 October 2020.
122. "Russia introduces Su-34 into fray" (<https://www.janes.com/defence-news/defence/latest/ukraine-conflict-russia-introduces-su-34-into-fray>). Gareth Jennings. 1 March 2022. Archived (<https://web.archive.org/web/20220301162511/https://www.janes.com/defence-news/defence/latest/ukraine-conflict-russia-introduces-su-34-into-fray>) from the original on 1 March 2022. Retrieved 1 March 2022.
123. Axe, David. "Ukraine said to have mauled a Russian fighter regiment, shooting down a quarter of its crews" (<https://www.forbes.com/sites/davidaxe/2022/09/29/ukrainian-air-defenses-mauled-a-russian-fighter-regiment-shooting-down-a-quarter-of-its-crews/>). *Forbes*. Archived (<https://web.archive.org/web/20220930013136/https://www.forbes.com/sites/davidaxe/2022/09/29/ukrainian-air-defenses-mauled-a-russian-fighter-regiment-shooting-down-a-quarter-of-its-crews/>) from the original on 30 September 2022. Retrieved 30 September 2022.

124. "Attack on Europe: Documenting Russian Equipment Losses During the 2022 Russian Invasion of Ukraine" (<https://www.oryxspioenkop.com/2022/02/attack-on-europe-documenting-equipment.html>). Archived (<https://web.archive.org/web/20220604015103/https://www.oryxspioenkop.com/2022/02/attack-on-europe-documenting-equipment.html>) from the original on 4 June 2022. Retrieved 5 January 2023.
125. Axe, David. "Russia's Best Su-34 Fighter-Bombers Are Falling From The Sky In Startling Numbers" (<https://www.forbes.com/sites/davidaxe/2023/12/24/russias-best-su-34-fighter-bombers-are-falling-from-the-sky-in-startling-numbers/>). *Forbes*. Retrieved 3 June 2024.
126. "The Mysterious Case of the Missing Russian Air Force" (<https://rusi.orghttps://rusi.org>). *rusi.orghttps*. Retrieved 3 June 2024.
127. "What did Ukraine use to shoot down 13 russian aircraft in two weeks and how many of them did the enemy have left?" (<https://visitukraine.today/blog/3467/what-did-ukraine-use-to-shoot-down-13-russian-aircraft-in-two-weeks-and-how-many-of-them-did-the-enemy-have-left>). *visitukraine.today*. Retrieved 3 June 2024.
128. "War-wrecked Russian arms lose appeal among Southeast Asia buyers: analysts" (<https://www.scmp.com/week-asia/politics/article/3177545/russian-military-hardware-wrecked-ukraine-war-could-lower-demand>). *South China Morning Post*. 13 May 2022. Retrieved 3 June 2024.
129. Oryx. "Attack On Europe: Documenting Russian Equipment Losses During The Russian Invasion Of Ukraine" (<https://www.oryxspioenkop.com/2022/02/attack-on-europe-documenting-equipment.html>). *Oryx*. Retrieved 6 July 2025.
130. "ASN Wikibase Occurrence # 276094" (<https://aviation-safety.net/wikibase/276094>). *Aviation Safety Network*. Archived (<https://web.archive.org/web/20220828131800/https://aviation-safety.net/wikibase/276094>) from the original on 28 August 2022. Retrieved 30 October 2022.
131. Operator Starsky; Denys Davydov (2 July 2022), *Unexpected find in Buzova: exploring the wreck of the russian bomber with @Denys Davydov* (<https://www.youtube.com/watch?v=Md1DKA7Lglg>), archived (<https://web.archive.org/web/20221118134336/https://www.youtube.com/watch?v=Md1DKA7Lglg>) from the original on 18 November 2022, retrieved 18 November 2022
132. 28.2.22 Су34 «5кр»RF81259 2САП Шагол Челябинск,вкс рф . Сбит «БукМ1»ВСУ  Бuzова,Киевская (https://www.youtube.com/watch?v=_hdKVUDYzmk), 30 October 2022, archived (https://web.archive.org/web/20221118134340/https://www.youtube.com/watch?v=_hdKVUDYzmk) from the original on 18 November 2022, retrieved 18 November 2022
133. Benassatto, Leonardo (1 May 2022). "Ukrainian forges plane wreckage into key fobs to fund war effort" (<https://www.reuters.com/business/aerospace-defense/ukrainian-forges-plane-wreckage-into-key-fobs-fund-war-effort-2022-04-30/>). *Reuters*. Archived (<https://web.archive.org/web/20220430211724/https://www.reuters.com/business/aerospace-defense/ukrainian-forges-plane-wreckage-into-key-fobs-fund-war-effort-2022-04-30/>) from the original on 30 April 2022. "The aircraft is a Russian Su-34 two-seater tactical fighter-bomber that the Ukrainian military says it shot down over the town of Borodianka, northwest of Kyiv, early in March... Ukraine's defence ministry has posted images of the wreckage, which it says had the tail number RF-81251 and the call sign "31 Red"."
134. "На Київщині ідентифікували збитий російський Су-34" (<https://mil.in.ua/uk/news/na-kyivshhyni-identyfikuvaly-zbytyj-rosijskyj-su-34/>). *mil.in.ua* (in Ukrainian). 17 April 2022. Archived (<https://web.archive.org/web/20220417130046/https://mil.in.ua/uk/news/na-kyivshhyni-identyfikuvaly-zbytyj-rosijskyj-su-34/>) from the original on 17 April 2022. Retrieved 5 May 2022.
135. "Accident Sukhoi Su-34 RF-81251 31 R" (<https://aviation-safety.net/wikibase/276063>). Archived (<https://web.archive.org/web/20230301182212/https://aviation-safety.net/wikibase/276063>) from the original on 1 March 2023. Retrieved 3 March 2024.
136. "ASN Occurrence # 276187" (<https://aviation-safety.net/wikibase/276187>). *Aviation Safety Network*. 5 March 2021. Archived (<https://web.archive.org/web/20220306022005/https://aviation-safety.net/wikibase/276187>) from the original on 6 March 2022. Retrieved 5 May 2022.

137. "Two Russian aircraft shot down on Saturday – Arestovych" (<https://en.interfax.com.ua/news/general/807703.html>). *Interfax Ukraine*. 5 March 2022. Archived (<https://web.archive.org/web/20220306141723/https://en.interfax.com.ua/news/general/807703.html>) from the original on 6 March 2022. Retrieved 5 May 2022.
138. Hopkins, Valerie; Varenikova, Maria (30 May 2022). "A Russian Plane Crashed into a House. Death Was Parceled Out Randomly" (<https://www.nytimes.com/2022/05/30/world/europe/ukraine-russia-chernihiv.html>). *The New York Times*. Archived (<https://web.archive.org/web/20221030201009/https://www.nytimes.com/2022/05/30/world/europe/ukraine-russia-chernihiv.html>) from the original on 30 October 2022. Retrieved 30 May 2022.
139. "Accident Sukhoi Su-34 RF-95070 06 R" (<https://aviation-safety.net/wikibase/276246>). Archived (<https://web.archive.org/web/20230822093444/https://aviation-safety.net/wikibase/276246>) from the original on 22 August 2023. Retrieved 3 March 2024.
140. "У Харкові знайшли уламки збитого Су-34 Росії" (<https://mil.in.ua/uk/news/u-harkovi-znajshly-ulamky-zbytogo-su-34-rosiyi/>) (in Ukrainian). 7 March 2022. Archived (<https://web.archive.org/web/20220307144937/https://mil.in.ua/uk/news/u-harkovi-znajshly-ulamky-zbytogo-su-34-rosiyi/>) from the original on 7 March 2022. Retrieved 11 April 2022.
141. "Incident Sukhoi Su-34" (<https://aviation-safety.net/wikibase/302618>). Archived (<https://web.archive.org/web/20240303144521/https://aviation-safety.net/wikibase/302618>) from the original on 3 March 2024. Retrieved 3 March 2024.
142. "В Україні ідентифікували п'ятий збитий Су-34 Росії" (<https://mil.in.ua/uk/news/v-ukrayini-identyfikovali-p-yatyj-zbytij-su-34-rosiyi/>) (in Ukrainian). 10 April 2022.
143. "ASN Wikibase Occurrence # 277276" (<https://aviation-safety.net/wikibase/277276>). *Aviation Safety Network*. Archived (<https://web.archive.org/web/20220410084754/https://aviation-safety.net/wikibase/277276>) from the original on 10 April 2022. Retrieved 5 May 2022.
144. "Incident Sukhoi Su-34 RF-95858 43 R" (<https://aviation-safety.net/wikibase/277678>). Archived (<https://web.archive.org/web/20240303144521/https://aviation-safety.net/wikibase/277678>) from the original on 3 March 2024. Retrieved 3 March 2024.
145. "На Запоріжжі виявили збитий російський винищувач Су-34" (<https://mil.in.ua/uk/news/na-zaporizhzh-vyavyly-zbytij-rosijskyj-vynyshhuvach-su-34/>). *mil.in.ua* (in Ukrainian). 26 April 2022. Archived (<https://web.archive.org/web/20220426202326/https://mil.in.ua/uk/news/na-zaporizhzh-vyavyly-zbytij-rosijskyj-vynyshhuvach-su-34/>) from the original on 26 April 2022. Retrieved 5 May 2022.
146. Newdick, Thomas (25 April 2022). "Ukraine Situation Report: Top U.S. Officials Pledge 'Advanced Weapons' During Kyiv Visit" (<https://www.thedrive.com/the-war-zone/ukraine-situation-report-top-u-s-officials-pledge-advanced-weapons-during-kyiv-visit>). *The Drive*. Archived (<https://web.archive.org/web/20220426003424/https://www.thedrive.com/the-war-zone/ukraine-situation-report-top-u-s-officials-pledge-advanced-weapons-during-kyiv-visit>) from the original on 26 April 2022. Retrieved 5 May 2022. "...footage has also emerged showing the apparent demise of a Russian Aerospace Forces Su-34 Fullback strike aircraft. Although the location and date of the incident remain unconfirmed, the aircraft is clearly seen in a flat spin, in which an aircraft plummets from the sky in rotational motion with no forward speed. A flat spin is sometimes impossible to recover from, which was the case here, although it's unclear whether the jet was hit by an air defense system or suffered some other kind of failure."
147. "На Харківщині збили російський винищувач типу Су" (<https://mil.in.ua/uk/news/na-harkivshhyni-zbyly-rosijskyj-vynyshhuvach-typu-su/>). *mil.in.ua* (in Ukrainian). 25 April 2022. Archived (<https://web.archive.org/web/20220425120238/https://mil.in.ua/uk/news/na-harkivshhyni-zbyly-rosijskyj-vynyshhuvach-typu-su/>) from the original on 25 April 2022. Retrieved 5 May 2022.
148. "Accident Sukhoi Su-34 RF-95846 16 R" (<https://aviation-safety.net/wikibase/278252>). Archived (<https://web.archive.org/web/20230822093445/https://aviation-safety.net/wikibase/278252>) from the original on 22 August 2023. Retrieved 3 March 2024.

149. Axe, David (18 July 2022). "The Russians Just Shot Down One Of Their Own Best Jets" (<https://www.forbes.com/sites/davidaxe/2022/07/18/the-russians-just-shot-down-one-of-their-own-best-jets/>). *Forbes*. Archived (<https://web.archive.org/web/20221206112217/https://www.forbes.com/sites/davidaxe/2022/07/18/the-russians-just-shot-down-one-of-their-own-best-jets/>) from the original on 6 December 2022. Retrieved 15 September 2022.
150. "Russia Accidentally Shoots Down Their Own \$36M Su-34 Bomber, Ukraine Claims" (<https://www.newsweek.com/russia-shoots-su-34-bomber-luhansk-ukraine-claims-1725524>). *Newsweek*. 18 July 2022. Archived (<https://web.archive.org/web/20220718171116/https://www.newsweek.com/russia-shoots-su-34-bomber-luhansk-ukraine-claims-1725524>) from the original on 18 July 2022. Retrieved 18 July 2022.
151. "Incident Sukhoi Su-34M RF-95890 51 R" (<https://aviation-safety.net/wikibase/280487>). Archived (<https://web.archive.org/web/20220811151715/https://aviation-safety.net/wikibase/280487>) from the original on 11 August 2022. Retrieved 3 March 2024.
152. "Incident Sukhoi Su-34 RF-95004 20 R" (<https://aviation-safety.net/wikibase/282736>). Archived (<https://web.archive.org/web/20220912091609/https://aviation-safety.net/wikibase/282736>) from the original on 12 September 2022. Retrieved 3 March 2024.
153. Malyasov, Dylan (11 September 2022). "Ukraine says its military shot down Russian Su-34 strike aircraft" (<https://defence-blog.com/ukraine-says-its-military-shot-down-russian-su-34-strike-aircraft/>). Archived (<https://web.archive.org/web/20220911063534/https://defence-blog.com/ukraine-says-its-military-shot-down-russian-su-34-strike-aircraft/>) from the original on 11 September 2022. Retrieved 11 September 2022.
154. "Incident Sukhoi Su-34" (<https://aviation-safety.net/wikibase/282749>). Archived (<https://web.archive.org/web/20240228060529/https://aviation-safety.net/wikibase/282749>) from the original on 28 February 2024. Retrieved 3 March 2024.
155. "Incident Sukhoi Su-34 RF-95005 22 R" (<https://aviation-safety.net/wikibase/283707>). Archived (<https://web.archive.org/web/20220925103743/https://aviation-safety.net/wikibase/283707>) from the original on 25 September 2022. Retrieved 3 March 2024.
156. "Incident Sukhoi Su-34 RF-81852 09 R" (<https://aviation-safety.net/wikibase/285879>). Archived (<https://web.archive.org/web/20230718105201/https://aviation-safety.net/wikibase/285879>) from the original on 18 July 2023. Retrieved 3 March 2024.
157. "Accident Sukhoi Su-34" (<https://aviation-safety.net/wikibase/308871>). Archived (<https://web.archive.org/web/20240228083923/https://aviation-safety.net/wikibase/308871>) from the original on 28 February 2024. Retrieved 3 March 2024.
158. "Accident Sukhoi Su-34" (<https://aviation-safety.net/wikibase/312488>). Archived (<https://web.archive.org/web/20240229224422/https://aviation-safety.net/wikibase/312488>) from the original on 29 February 2024. Retrieved 3 March 2024.
159. "Incident Sukhoi Su-34" (<https://aviation-safety.net/wikibase/348942>). Archived (<https://web.archive.org/web/20240303144522/https://aviation-safety.net/wikibase/348942>) from the original on 3 March 2024. Retrieved 3 March 2024.
160. "Ukraine Says It Shot Down Three Russian Su-34 Fullbacks" (<https://www.twz.com/ukraine-says-it-shot-down-three-russian-su-34-fullbacks>). *twz.com*. 22 December 2023. Archived (<https://web.archive.org/web/20240303144521/https://www.twz.com/ukraine-says-it-shot-down-three-russian-su-34-fullbacks>) from the original on 3 March 2024. Retrieved 3 March 2024.
161. "Ukraine claims to down three Russian fighter-bomber jets" (<https://www.reuters.com/world/europe/ukraine-claims-down-three-russian-fighter-bomber-jets-2023-12-23/>). *Reuters*. 23 December 2023. Retrieved 23 December 2023.

162. David Axe (4 January 2024). "A Ukrainian Saboteur Traveled 900 Miles to A Snowy Russian Airfield And, In The Dead Of Night, Lit A Russian Sukhoi Fighter-Bomber On Fire" (<https://www.forbes.com/sites/davidaxe/2024/01/04/a-ukrainian-saboteur-traveled-600-miles-to-a-snowy-russian-airfield-and-in-the-dead-of-night-lit-a-russian-sukhoi-fighter-bomber-on-fire/?sh=3453204c3d6e>). *Forbes*. Archived (<https://web.archive.org/web/20240126173726/https://www.w.forbes.com/sites/davidaxe/2024/01/04/a-ukrainian-saboteur-traveled-600-miles-to-a-snowy-russian-airfield-and-in-the-dead-of-night-lit-a-russian-sukhoi-fighter-bomber-on-fire/?sh=3453204c3d6e>) from the original on 26 January 2024. Retrieved 5 January 2024.
163. "Incident Sukhoi Su-34" (<https://aviation-safety.net/wikibase/349609>). Archived (<https://web.archive.org/web/20240303144522/https://aviation-safety.net/wikibase/349609>) from the original on 3 March 2024. Retrieved 3 March 2024.
164. "Ukraine Downs Russian Su-34 Fighter Jet Over Luhansk Region" (<https://www.kyivpost.com/post/27357>). Kyiv Post. 30 January 2024. Archived (<https://web.archive.org/web/20240131122057/https://www.kyivpost.com/post/27357>) from the original on 31 January 2024. Retrieved 30 January 2024.
165. "Air Force: Ukraine downs Russian Su-34 aircraft" (<https://kyivindependent.com/air-force-ukraine-downs-russian-su-34-aircraft/>). *Kyiv Independent*. 27 February 2024. Archived (<https://web.archive.org/web/20240227095916/https://kyivindependent.com/air-force-ukraine-downs-russian-su-34-aircraft/>) from the original on 27 February 2024. Retrieved 27 February 2024.
166. "Ukraine's Defence Forces down 2 more Russian Su-34 jets: three jets downed on 29 February" (<https://www.pravda.com.ua/eng/news/2024/02/29/7444287/>). *Ukrainska Pravda*. 29 February 2024. Archived (<https://web.archive.org/web/20240229111552/https://www.pravda.com.ua/eng/news/2024/02/29/7444287/>) from the original on 29 February 2024. Retrieved 29 February 2024.

"February is the shortest month of the year, but our sky defenders have achieved the greatest results in downing russian jets since October 2022" (<https://twitter.com/DefenceU/status/1763195543885324370>). *Ministry of Defense of Ukraine via Twitter*. 29 February 2024. Archived (<https://web.archive.org/web/20240229150837/https://twitter.com/DefenceU/status/1763195543885324370>) from the original on 29 February 2024. Retrieved 29 February 2024.
167. Julia Struck (29 February 2024). "Ukrainian Air Force Downs Three More Russian Fighter-Bombers, Bringing Total to 12 in 13 Days" (<https://www.kyivpost.com/post/28805>). Kyiv Post. Archived (<https://web.archive.org/web/20240229132517/https://www.kyivpost.com/post/28805>) from the original on 29 February 2024. Retrieved 29 February 2024.
168. "Ukraine Shoots Down Russian Su-34 Fighter Jet, Says Air Force Commander" (<https://www.rferl.org/a/ukraine-shoots-down-russia-su034-jet/32837137.html>). *Radiofreeeurope/Radioliberty*. Archived (<https://web.archive.org/web/20240303153027/https://www.rferl.org/a/ukraine-shoots-down-russia-su034-jet/32837137.html>) from the original on 3 March 2024. Retrieved 3 March 2024.
169. YURI ZORIA (15 June 2024). "At least two Russian Su-34 bombers damaged in Ukrainian drone attack at Morozovsk airfield' " (<https://euromaidanpress.com/2024/06/15/at-least-two-russian-su-34-bombers-damaged-in-ukrainian-drone-attack-at-morozovsk-airfield/>). Euromaidanpress. Retrieved 16 June 2024.
170. Mikhaila Friel (14 August 2024). "Ukraine said it shot down a \$36 million Su-34 bomber jet inside Russia" (<https://www.businessinsider.com/ukraine-shot-down-russia-su-34-jet-over-ukrsk-military-2024-8>). *Business Insider*. Retrieved 14 August 2024.
171. "Ukraine war latest: Ukrainian drones reportedly strike 4 fighter jets in Russia" (<https://kyivindependent.com/ukraine-war-latest-ukraines-long-range-drones-attack-airfield-in-russias-volgograd-oblast-hitting-4-fighter-jets/>). Kyiv: The Kyiv Independent. 27 June 2025. Retrieved 4 July 2025.

172. "Ukraine shoots down Russian Su-34 bomber over Zaporizhzhia Oblast, Air Force says" (<https://kyivindependent.com/ukraine-shoots-down-russian-su-34-bomber-over-zaporizhzhia-oblast-air-force-says/>). *The Kyiv Independent*. 25 September 2025. Retrieved 25 September 2025.
173. "На вологодские ледяные заторы сбросили четыре тонны бомб" (<http://rg.ru/2016/04/18/reg-szfo/na-vologodskie-lediane-zatory-sbrosili-chetyre-tonny-bomb.html>). *Rg.ru*. 18 April 2016. Archived (<https://web.archive.org/web/20160523025428/http://rg.ru/2016/04/18/reg-szfo/na-vologodskie-lediane-zatory-sbrosili-chetyre-tonny-bomb.html>) from the original on 23 May 2016. Retrieved 1 July 2017.
174. "Су-34 разбомбили Кобылинские перекаты, скованные льдом" (<http://www.vesti.ru/doc.html?id=2744423>). *Vesti.ru*. 19 May 2017. Archived (<https://web.archive.org/web/20170201173154/http://www.vesti.ru/doc.html?id=2744423>) from the original on 1 February 2017. Retrieved 1 July 2017.
175. "Экипажи бомбардировщиков Су-34 ЗВО выполнили бомбометание по ледяным заторам на реке Сухона в Вологодской области: Министерство обороны Российской Федерации" (http://function.mil.ru/news_page/country/more.htm?id=12083379@egNews). *Function.mil.ru*. 18 April 2016. Archived (https://web.archive.org/web/20160428161928/http://function.mil.ru/news_page/country/more.htm?id=12083379@egNews) from the original on 28 April 2016. Retrieved 1 July 2017.
176. Insinna, Valerie (5 January 2016). "Algeria Orders 12 Su-34 'Fullback' Fighter-Bombers from Russia" (<http://www.defensenews.com/story/defense/2016/01/05/algeria-orders-12-su-34-fullback-fighter-bombers-russia/78319912/>). *defensenews.com*. Archived (<https://web.archive.org/web/20240312192531/https://www.defensenews.com/home/2016/01/05/algeria-orders-12-su-34-fullback-fighter-bombers-from-russia/>) from the original on 12 March 2024. Retrieved 31 December 2019.
177. "Russia, Algeria To Sign Deal for Su-34 Sale by End of 2016" (<http://www.defensenews.com/story/defense/air-space/2016/03/02/algeria-sukhoi-su-34-sale-russia/81218660/>). *defensenews.com*. 2 March 2016. Archived (<https://web.archive.org/web/20240312192540/https://www.defensenews.com/air/2016/03/02/russia-algeria-to-sign-deal-for-su-34-sale-by-end-of-2016/>) from the original on 12 March 2024. Retrieved 31 December 2019.
178. "Algeria restarts negotiations [sic] on Su 32" (<https://www.menadefense.net/algeria/algeria-restarts-negotiations-on-su-32/>). *menadefense.net*. 22 July 2017. Archived (<https://web.archive.org/web/20191222081244/https://www.menadefense.net/algeria/algeria-restarts-negotiations-on-su-32/>) from the original on 22 December 2019. Retrieved 31 December 2019.
179. Cooper, Tom (24 October 2016). "The Algerian Su-34 Is a Myth" (<https://medium.com/war-is-boring/the-algerian-su-34-is-a-myth-dba320792554>). *War is Boring*. Archived (<https://web.archive.org/web/20200306161321/https://medium.com/war-is-boring/the-algerian-su-34-is-a-myth-dba320792554>) from the original on 6 March 2020. Retrieved 7 January 2020.
180. "L'Algérie serait le premier client export pour le Su-57 et le Su-34" (<https://www.menadefense.net/algerie/lalgerie-serait-le-premier-client-export-pour-le-su-57-et-le-su-34/>). *menadefense.net*. 27 December 2019. Archived (<https://web.archive.org/web/20191230085917/https://www.menadefense.net/algerie/lalgerie-serait-le-premier-client-export-pour-le-su-57-et-le-su-34/>) from the original on 30 December 2019. Retrieved 31 December 2019.
181. "Началось производство первой партии Су-34 для Алжира" (<https://xn--b1aga5aadd.xn--p1ai/2021/%D0%A2%D0%BE%D1%80%D0%B3%D0%BE%D0%B2%D0%BB%D1%8F%D0%9E%D1%80%D1%83%D0%B6%D0%B8%D0%B5%D0%BC16/>) [Production of the first batch of Su-34 for Algeria began]. *Mil.press* (in Russian). 15 March 2021. Archived (<https://web.archive.org/web/20240125183642/https://xn--b1aga5aadd.xn--p1ai/2021/%D0%A2%D0%BE%D1%80%D0%B3%D0%BE%D0%B2%D0%BB%D1%8F%D0%9E%D1%80%D1%83%D0%B6%D0%B8%D0%B5%D0%BC16/>) from the original on 25 January 2024. Retrieved 25 December 2023.
182. "Russia denies supplying Algeria with Su-34 fighters" (<https://www.middleeastmonitor.com/20210317-russia-denies-supplying-algeria-with-su-34-fighters/>).

183. Nikolov, Boyko (20 May 2025). "Russian Su-34 in desert camo hints at export deal unveiled" (<https://bulgarianmilitary.com/2025/05/20/russian-su-34-in-desert-camo-hints-at-export-deal-unveiled/>). *Bulgarian Military Industry Review*. Archived (<https://web.archive.org/web/20250923044042/https://bulgarianmilitary.com/2025/05/20/russian-su-34-in-desert-camo-hints-at-export-deal-unveiled/>) from the original on 23 September 2025. Retrieved 18 July 2025.
184. "Su-32." Rosoboronexport, Archive (<https://web.archive.org/web/20170805084229/http://roe.ru/eng/catalog/aerospace-systems/bombers/su-32/>)
185. "Su-34E." Rosoboronexport (<https://roe.ru/en/production/aviatsionnaya-tehnika-i-vooruzhenie-vozdushno-kosmicheskikh-sil/samolety/mnogotselevye-istrebiteli-i-istrebiteli-bombardirovshchiki/su-34e/?theme=theme-lightblue%20Su-34E>)
186. "The source learned about the new contract for the supply of a batch of Su-34 for the Ministry of Defense. (In Russian)" (<https://www.interfax.ru/russia/712576>). Interfax. 10 June 2020. Archived (<https://web.archive.org/web/20210515202801/https://www.interfax.ru/russia/712576>) from the original on 15 May 2021. Retrieved 15 May 2021.
187. "The Aerospace Forces began to receive updated Su-34 NVO bombers (In Russian)" (<https://aviation21.ru/vks-nachali-poluchat-obnovlyonnye-bombardirovshchiki-su-34-nvo/>). *Aviation of Russia*. 19 March 2021. Archived (<https://web.archive.org/web/20210515202801/https://aviation21.ru/vks-nachali-poluchat-obnovlyonnye-bombardirovshchiki-su-34-nvo/>) from the original on 15 May 2021. Retrieved 15 May 2021.
188. "Russian Su-34 bombers received new reconnaissance systems (In Russian)" (<https://rg.ru/2021/04/12/rossijskie-bombardirovshchiki-su-34-poluchili-novye-kompleksy-razvedki.html>). *RG*. 12 April 2021. Archived (<https://web.archive.org/web/20210515204303/https://rg.ru/2021/04/12/rossijskie-bombardirovshchiki-su-34-poluchili-novye-kompleksy-razvedki.html>) from the original on 15 May 2021. Retrieved 15 May 2021.
189. "Су-34 впервые применили в зоне СВО крылатые ракеты большой дальности" (<https://ria.ru/20230915/su-34-1896437788.html>). 15 September 2023. Archived (<https://web.archive.org/web/20230916131807/https://ria.ru/20230915/su-34-1896437788.html>) from the original on 16 September 2023. Retrieved 16 September 2023.
190. Kohanets, Roman (11 July 2025). "Russia Upgrades Su-34 With Sych Recon Pods for Deep-Strike Surveillance in Ukraine" (<https://united24media.com/latest-news/russia-upgrade-s-su-34-with-sych-recon-pods-for-deep-strike-surveillance-in-ukraine-9811>). *UNITED24 Media*. Retrieved 18 July 2025.
191. "Еще пять Су-34 прибыли в Балтимор" (<https://bmpd.livejournal.com/443923.html>). *bmpd.livejournal.com*. 26 January 2013. Archived (<https://web.archive.org/web/20191116163743/https://bmpd.livejournal.com/443923.html>) from the original on 16 November 2019. Retrieved 16 November 2019.
192. "ВКС России получили еще четыре фронтовых бомбардировщика Су-34" (<https://bmpd.livejournal.com/2974493.html>). *bmpd.livejournal.com*. 28 November 2017. Archived (<https://web.archive.org/web/20190404194524/https://bmpd.livejournal.com/2974493.html>) from the original on 4 April 2019. Retrieved 16 November 2019.
193. "Телерепортаж о Су-34 из Морозовска" (<https://bmpd.livejournal.com/1694294.html>). *bmpd.livejournal.com*. 23 January 2016. Archived (<https://web.archive.org/web/20191116162235/https://bmpd.livejournal.com/1694294.html>) from the original on 16 November 2019. Retrieved 16 November 2019.
194. "Авиационный полк в Шаголе завершил перевооружение на фронтовые бомбардировщики Су-34" (<https://bmpd.livejournal.com/3795166.html>). *bmpd.livejournal.com*. 3 October 2019. Archived (<https://web.archive.org/web/20191006084034/https://bmpd.livejournal.com/3795166.html>) from the original on 6 October 2019. Retrieved 16 November 2019.
195. "Су-34 в Липецке" (<https://bmpd.livejournal.com/190389.html>). *bmpd.livejournal.com*. 30 March 2012. Archived (<https://web.archive.org/web/20191116163742/https://bmpd.livejournal.com/190389.html>) from the original on 16 November 2019. Retrieved 16 November 2019.

196. "Два новых ударных самолета Су-34 пополнили боевой состав Липецкого авиацентра: Министерство обороны Российской Федерации" (http://function.mil.ru/news_page/country/more.htm?id=12075630@egNews). *function.mil.ru*. 29 January 2016. Archived (https://web.archive.org/web/20200627081653/https://function.mil.ru/news_page/country/more.htm?id=12075630@egNews) from the original on 27 June 2020. Retrieved 16 November 2019.
197. "Впервые с новой взлетно-посадочной полосы Ахтубинска взлетели боевые самолеты BBC России" (<https://tass.ru/politika/528306>). TASS. 7 May 2013. Archived (<https://web.archive.org/web/20191119151753/https://tass.ru/politika/528306>) from the original on 19 November 2019. Retrieved 16 November 2019.
198. "Russia increases operational jets in Syria after US missile strike [+ Photos]" (<https://web.archive.org/web/20180502012315/https://www.almasdarnews.com/article/russia-increases-operational-jets-in-syria-after-us-missile-strike-photos/>). *Al-Masdar News*. 30 April 2018. Archived from the original (<https://www.almasdarnews.com/article/russia-increases-operational-jets-in-syria-after-us-missile-strike-photos/>) on 2 May 2018. Retrieved 16 November 2019.
199. "Algeria to Receive First Su-34 Fighter Bomber From Russia After Six-Year Wait" (https://en.defence-ua.com/news/algeria_to_receive_first_su_34_fighter_bomber_from_russia_after_six_year_wait-15476.html). *en.defence-ua.com*. 21 December 2025. Retrieved 21 December 2025.
200. Carlin, Maya (10 September 2025). "Algeria to Receive Russian Su-34 Fighters" (<https://nationalinterest.org/blog/buzz/algeria-to-receive-russian-su-34-fighters-mc-090925>). *The National Interest*. Retrieved 26 May 2026.
201. "Algeria has become the first foreign operator of Russian Su-34 fighter jets" (<https://ua.news/en/world/alzhir-stav-pershim-inozemnim-operatorom-rosiiskikh-vinishchuvachiv-su-34>). *UA.NEWS*. 26 May 2026. Retrieved 26 May 2026.
202. "Su-32: Aircraft performance" (<https://web.archive.org/web/20151118150836/http://www.sukhoi.org/eng/planes/military/su32/lth/>). *Sukhoi*. Archived from the original (<http://www.sukhoi.org/eng/planes/military/su32/lth/>) on 18 November 2015. Retrieved 25 November 2015., "Armaments" (<https://web.archive.org/web/20111111231639/http://sukhoi.org/eng/planes/military/su32/arms/>). *Sukhoi*. Archived from the original (<http://www.sukhoi.org/eng/planes/military/su32/arms/>) on 11 November 2011. Retrieved 4 July 2011.
203. Gordon & Davison (2006), pp. 92, 95–96.
204. "Su-34 (Su-27IB) Flanker Fighter Bomber Aircraft, Russia" (<https://www.airforce-technology.com/projects/su34>). *airforce-technology.com*. Archived (<https://web.archive.org/web/20120923124328/http://www.airforce-technology.com/projects/su34>) from the original on 23 September 2012. Retrieved 5 October 2012.
205. "SU-34" (<http://airwar.ru/image/idop/bomber/su34/su34-2.gif>). *airwar.ru*. Archived (<https://web.archive.org/web/20110928102422/http://airwar.ru/image/idop/bomber/su34/su34-2.gif>) from the original on 28 September 2011. Retrieved 5 October 2012.
206. "Бондарев: Су-34 гораздо лучше предшественников" (http://ria.ru/defense_safety/20131226/986658004.html) [Bondarev: Su-34 is much better than predecessors]. *RIA Novosti* (in Russian). 26 December 2013. Archived (https://web.archive.org/web/20141025045309/http://ria.ru/defense_safety/20131226/986658004.html) from the original on 25 October 2014. Retrieved 14 November 2014.
207. "Su-34" (http://www.deagel.com/Combat-Aircraft/Su-34_a000181001.aspx). *Deagel.com*. Archived (https://web.archive.org/web/20170629064738/http://www.deagel.com/Combat-Aircraft/Su-34_a000181001.aspx) from the original on 29 June 2017. Retrieved 1 July 2017.
208. Lednicer, David. "The Incomplete Guide to Airfoil Usage" (<https://m-selig.ae.illinois.edu/ads/aircraft.html>). *m-selig.ae.illinois.edu*. Archived (<https://web.archive.org/web/20130905115027/https://m-selig.ae.illinois.edu/ads/aircraft.html>) from the original on 5 September 2013. Retrieved 16 April 2019.

209. "Su-34 Fullback" (<https://web.archive.org/web/20150118085207/http://defense-update.com/products/s/su-34.htm>). *Defense-update.com*. Archived from the original (<http://defense-update.com/products/s/su-34.htm>) on 18 January 2015. Retrieved 1 March 2012.
210. "Why is Russia's Su-34 Fighter-Bomber Failing in Ukraine?" (<https://www.19fortyfive.com/2022/05/su-34-failures-ukraine/>). 11 May 2022. Archived (<https://web.archive.org/web/20220615005508/https://www.19fortyfive.com/2022/05/su-34-failures-ukraine/>) from the original on 15 June 2022. Retrieved 23 June 2022.
211. "Бондарев: Су-34 гораздо лучше предшественников" (<https://ria.ru/20131226/986658004.html>). 26 December 2013. Archived (<https://web.archive.org/web/20190903104518/https://ria.ru/20131226/986658004.html>) from the original on 3 September 2019. Retrieved 3 September 2019.
212. "На Дальнем Востоке Су-34 на учениях взлетели на высоту 17 километров" (<https://ria.ru/20200207/1564381945.html>). 7 February 2020. Archived (<https://web.archive.org/web/20200208162509/https://ria.ru/20200207/1564381945.html>) from the original on 8 February 2020. Retrieved 8 February 2020.
213. "Aircraft performance" (<https://web.archive.org/web/20151118150836/http://www.sukhoi.org/eng/planes/military/su32/lth/>). *Sukhoi*. Archived from the original (<http://www.sukhoi.org/eng/planes/military/su32/lth/>) on 18 November 2015. Retrieved 27 October 2015.
214. "Истребитель-бомбардировщик Су-34" (<http://structure.mil.ru/structure/forces/air/weapons/aviation/more.htm?id=10332831@morfMilitaryModel>) [Su-34 Fighter-bomber]. *structure.mil.ru* (in Russian). Archived (<https://web.archive.org/web/20150403162510/http://structure.mil.ru/structure/forces/air/weapons/aviation/more.htm?id=10332831@morfMilitaryModel>) from the original on 3 April 2015. Retrieved 7 April 2015.
215. "Su-34 Pilot Killed My Brother - Ukrainian Residents Recall Horror After Downing Russia's Top Fighter Bomber" (<https://eurasianimes.com/su-34-pilot-killed-my-brother-ukrainian-residents-recall-horror/>). 23 May 2022. Archived (<https://web.archive.org/web/20220605200609/https://eurasianimes.com/su-34-pilot-killed-my-brother-ukrainian-residents-recall-horror/>) from the original on 5 June 2022. Retrieved 5 June 2022.
216. "Russian Su-34 uses Kinzhal hypersonic missile in special op — official" (<https://tass.com/defense/1669183>). Archived (<https://web.archive.org/web/20230905185434/https://tass.com/defense/1669183>) from the original on 5 September 2023. Retrieved 16 September 2023.
217. "РАКЕТНЫЙ КОМПЛЕКС С ПРОТИВОКОРАБЕЛЬНОЙ РАКЕТОЙ "ОНИКС" ("ЯХОНТ"), THE MISSILE COMPLEX WITH ANTI-SHIP MISSILE "ONYX" ("YAKHONT")" (<http://bastion-karpenko.narod.ru/Yahont.html>). *narod.ru* (in Russian). Archived (<https://web.archive.org/web/20150419191436/http://bastion-karpenko.narod.ru/Yahont.html>) from the original on 19 April 2015. Retrieved 7 April 2015.
218. "ЦАМТО / / Фронтовые бомбардировщики Су-34 получили возможность нести крылатые ракеты большой дальности" (<https://armstrade.org/includes/periodics/news/2023/0915/094075273/detail.shtml>). *armstrade.org*. Archived (<https://web.archive.org/web/20231001221107/https://armstrade.org/includes/periodics/news/2023/0915/094075273/detail.shtml>) from the original on 1 October 2023. Retrieved 16 September 2023.
219. "Russian Air Force Unveils FAB-3000 Powerful Aerial Bomb Fitted with Guidance Kit Used in Ukraine" (https://armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/russian-air-force-unveils-fab-3000-powerful-aerial-bomb-fitted-with-guidance-kit-used-in-ukraine?utm_content=cmp-true).
220. Boyko, Nikolov (5 September 2024). "Su-34HBO bomber appeared with 120km-range D-30SN glide bomb" (<https://bulgarianmilitary.com/2024/09/05/su-34hbo-light-bomber-appeared-with-120km-range-d-30sn-glide-bomb/>). *bulgarianmilitary*. Retrieved 7 October 2024.

Bibliography

- Eden, Paul, ed. (2004). *The Encyclopedia of Modern Military Aircraft*. London: Amber Books.

ISBN 978-1-904687-84-9.

- Gordon, Yefim (1999). *Sukhoi Su-27 Flanker: Air Superiority Fighter*. London: Airline Publishing. ISBN 978-1-84037-029-4.
- Gordon, Yefim; Davison, Peter (2006). *Sukhoi Su-27 Flanker*. North branch, Minnesota: Specialty Press. ISBN 978-1-58007-091-1.
- Spick, Mike (2000). "The Flanker". *The Great Book of Modern Warplanes*. St. Paul, Minnesota: Motorbooks International. ISBN 978-0-7603-0893-6.
- Williams, Mel, ed. (2002). "Sukhoi 'Super Flankers' ". *Superfighters: The Next Generation of Combat Aircraft*. Norwalk, Connecticut: AIRtime Publishing Inc. ISBN 978-1-880588-53-6.
- Wilson, Stewart (2000). *Combat Aircraft since 1945*. Fyshwick, Australia: Aerospace Publications. ISBN 978-1-875671-50-2.

External links

- Official Sukhoi Su-34 webpage at Sukhoi (<https://www.sukhoi.org/products/samolety/254/>)
- Su-32 page on Rosoboronexport's site (<http://roe.ru/eng/catalog/aerospace-systems/bombers/su-32/>)

External videos

on *History Channel* *Official YouTube Channel*(in Russian)



"The best Sukhoi aircraft". Su-34 (https://www.youtube.com/watch?v=sLLbafKV_JI) on YouTube

Retrieved from "https://en.wikipedia.org/w/index.php?title=Sukhoi_Su-34&oldid=1357581327"